

SMITHIAN FOLD THEORY - ASTRONOMY AND COSMOLOGY PAPER 001

From One Sky to Cosmos

An Exact, Zero-Parameter and Machine-Closed Foundational Reconstruction of
Astronomy and Cosmology from Smithian Fold Theory

Ernos Labs

Open Source Science Platform and Knowledge Tree

Maria Smith

Independent researcher and founder, Ernos Labs

Maria.Smith.Sftoe@gmail.com

Version 1.0.0 - current-evidence closed, extension-open foundation

72 required laws - 18,432 exact candidates

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Abstract

Astronomy is often presented as a catalogue of distant things and cosmology as a fit among inherited equations. This reconstruction begins lower. From the One and the Fold, and only through already admitted mathematical, informational, computational, physical, chemical and Earth dependencies, it asks what must be retained for any statement about a source beyond the observer to be distinguishable, reconstructible and falsifiable. The result is a foundational branch spanning source geometry, object identity, radiation, motion, time, stars, planetary systems, interstellar matter, galaxies, compact objects, multimessenger records, expansion, early-universe evidence, dark-sector inference and scientific handoffs. Each law is zero-parameter in its proof layer: no fitted constant, target value, irrational or imaginary proof number, conventional continuum or semantic numerical zero selects a survivor. Dimensional observations remain exact source records outside that proof layer. This paper documents the entire forward chain, the candidate grammars, the controls, the empirical boundary, the adverse result, the independent certificates and the open continuation programme.

Authorship, access and why this work is public

Maria Smith produced this work outside credentialed institutional access. That fact is not offered as a claim of personal exception. It is an indictment of how many capable minds science has excluded or silenced when money, credentials, paywalls, prestige and institutional permission are mistaken for rigor. The work must stand or fall on its visible derivations, executable tests, preserved adverse evidence and reproducible certificates-not on the biography of the author and not on the authority of a gatekeeper.

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The paper is licensed **CC BY 4.0**. The platform code is licensed **Apache-2.0**. Maria Smith retains authorship, copyright and creative rights. Openness does not erase attribution, and attribution does not convert public science into private permission. Contact: **Maria.Smith.Sftoe@gmail.com**. Submissions and review: <https://discord.gg/ucwGryVxGr>. GitHub: <https://github.com/MettaMazza>.

The non-negotiable mathematical constitution

Every claim in this branch enters through one theorem-**There Is No Nothing**-and admitted Fold dependencies. There are no added axioms and no free or fitted parameters. Proof-state arithmetic is exact and finite. Conventional numerical zero is not an element of the proof carrier; where a user interface, catalogue or source writes $\emptyset$, it is retained as a typed absence or missingness marker unless the source explicitly proves another meaning. Negative, irrational and imaginary quantities are not admitted as proof scalars. Completed infinities and ungenerated continua do not enter candidate generation. External decimals, signed coordinates and catalogue conventions are held labels in the measurement record; they compare the law but cannot become hidden premises.

For every claim, eight binary dimensions are generated literally: carrier, boundary, relation, record, evidence, provenance, generality and extension. This produces $2^8 = 256$ finite forms. Exactly one choice on each axis preserves the required distinction. The full Cartesian product is enumerated, every form is decided, and the single all-preserving form is independently reconstructed. Depth-independent closure means that adding one lawful finite source, epoch, channel, object or population member preserves prior identities and adverse rows while appending its own trace. It does not mean that the empirical universe is exhausted or that later observations are forbidden.

End-to-end empirical constitution

The complete 72-claim inventory, all 18,432 candidates, all predicted structural labels and all falsification conditions were sealed before external source identities or outcomes were opened. Primary custodians were then registered by purpose. Each target remained inaccessible to the capability-closed prediction process until its derivation seal existed. The evaluator opened the registered source record afterward, compared the result, ran deliberately changed controls, and preserved every transport failure and unfavorable row. The sealed engine rejected any missing dependency, incomplete census, non-unique survivor, failed control,

changed source, failed independent reconstruction or empirical boundary violation.

The immutable engine seal is

sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a. The

verification-authority seal is

sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8. They identify the unchanged common admission route; they are not substitutes for the derivations below.

Dependency and ownership boundary

The dependency chain is: foundational theorem → exact mathematics → information and computation → universal Physics → chemical identity and conservation → Earth observation handoff → Astronomy observation and history. Physics retains propagation, gravity, matter and constants. Chemistry retains species identity and transformation. Earth retains Earth-specific conditions. Engineering retains instrument implementation. Astronomy retains source identity, source-observer geometry, celestial populations, cosmic chronology and the custody of observational evidence. This division prevents double counting and prevents a telescope, simulation or catalogue from becoming a law generator.

Primary source register

- **ESA-GAIA-ARCHIVE-001 - European Space Agency.** Gaia ESA Archive. Registered purpose: observation_geometry, object_identity_classification, radiation_motion_time, stellar_systems, galaxies_populations. Locator: <https://gea.esac.esa.int/archive/>.
- **NASA-EXOPLANET-ARCHIVE-001 - NASA Exoplanet Science Institute.** NASA Exoplanet Archive. Registered purpose: planetary_small_body, stellar_systems. Locator: <https://exoplanetarchive.ipac.caltech.edu/>.
- **STSCI-MAST-001 - Space Telescope Science Institute.** Mikulski Archive for Space Telescopes. Registered purpose: observation_geometry, radiation_motion_time, stellar_systems, interstellar_enrichment_formation, galaxies_populations. Locator: <https://archive.stsci.edu/>.
- **CDS-SIMBAD-001 - Centre de Donnees astronomiques de Strasbourg.** SIMBAD astronomical database. Registered purpose: object_identity_classification, observation_geometry. Locator: <https://simbad.cds.unistra.fr/simbad/>.
- **MPC-DATA-SERVICES-001 - International Astronomical Union Minor Planet Center.** MPC Data Services. Registered purpose: planetary_small_body, observation_geometry. Locator: <https://data.minorplanetcenter.net/>.
- **SPARC-BTFR-001 - SPARC collaboration / Case Western Reserve University.** SPARC baryonic Tully-Fisher data. Registered purpose: galaxies_populations. Locator: https://astroweb.case.edu/SPARC/BTFR_Lelli2019.mrt.
- **GWOSC-PUBLIC-001 - LIGO Virgo KAGRA collaborations.** Gravitational Wave Open Science Center. Registered purpose: compact_high_energy_multimessenger. Locator: <https://gwosc.org/about/>.
- **ICECUBE-PUBLIC-DATA-001 - IceCube Collaboration.** IceCube public data releases. Registered purpose: compact_high_energy_multimessenger. Locator: <https://icecube.wisc.edu/science/data-releases/>.
- **NASA-HEASARC-001 - NASA High Energy Astrophysics Science Archive Research Center.** HEASARC data archive. Registered purpose: compact_high_energy_multimessenger, stellar_systems, galaxies_populations. Locator: <https://heasarc.gsfc.nasa.gov/docs/archive.html>.
- **NASA-LAMBDA-001 - NASA Goddard Space Flight Center.** Legacy Archive for Microwave Background Data Analysis. Registered purpose: large_scale_expansion, early_background_abundance, dark_horizon_unobserved. Locator: <https://lambda.gsfc.nasa.gov/>.
- **DESI-DR1-001 - Dark Energy Spectroscopic Instrument Collaboration.** DESI public Data Release 1. Registered purpose: galaxies_populations, large_scale_expansion, dark_horizon_unobserved. Locator: <https://data.desi.lbl.gov/doc/>.
- **SDSS-DATA-001 - Sloan Digital Sky Survey.** SDSS public data. Registered purpose: object_identity_classification, radiation_motion_time, galaxies_populations, large_scale_expansion. Locator: https://www.sdss.org/dr19/data_access/.
- **NASA-IRSA-001 - NASA Infrared Science Archive.** IRSA archives. Registered purpose: stellar_systems, planetary_small_body, interstellar_enrichment_formation, galaxies_populations. Locator: <https://irsa.ipac.caltech.edu/frontpage/>.

- **NASA-JPL-HORIZONS-001 - NASA Jet Propulsion Laboratory.** JPL Horizons system. Registered purpose: planetary_small_body. Locator: <https://ssd.jpl.nasa.gov/horizons/>.
- **NASA-ADS-001 - NASA Astrophysics Data System.** NASA ADS literature and data links. Registered purpose: inference_prediction_handoffs. Locator: <https://ui.adsabs.harvard.edu/>.

Family results

Source, observer, signal and geometry

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Astronomical source** (SFT-ASTRO-SOURCE-001): An astronomical source is a bounded carrier whose state produces or alters a registered signal along a declared path.
- **Astronomical observer** (SFT-ASTRO-OBSERVER-001): An observer is a source-located site, platform, instrument and clock relation, not an unlocated view.
- **Astronomical signal** (SFT-ASTRO-SIGNAL-001): A signal retains source, path, band, arrival, detector response and quality status.
- **Angular position and reference frame** (SFT-ASTRO-ANGULAR-POSITION-001): Angular position is a direction in a named frame and epoch with motion and uncertainty retained.
- **Astronomical distance** (SFT-ASTRO-DISTANCE-001): Distance is a method-bound source-observer relation whose ladder, calibration and range remain explicit.
- **Survey selection and non-detection** (SFT-ASTRO-SELECTION-001): Selection, detection, upper limit, non-detection, missing and outside-footprint status remain distinct.

Object identity, classification and catalogues

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Astronomical object identity** (SFT-ASTRO-OBJECT-001): Object identity is a reconstructible association of observations across epochs and channels.
- **Astronomical classification** (SFT-ASTRO-CLASSIFICATION-001): Classification retains measured features, method, alternatives and taxonomy version.
- **Astronomical population** (SFT-ASTRO-POPULATION-001): A population is a selection-bounded set with membership, volume, time, weights and censoring retained.
- **Astronomical morphology** (SFT-ASTRO-MORPHOLOGY-001): Morphology is a resolution- and band-bound organization of distinguishable components.
- **Astronomical composition inference** (SFT-ASTRO-COMPOSITION-001): Composition inference retains species evidence, excitation, method and alternatives.
- **Astronomical catalogue** (SFT-ASTRO-CATALOG-001): A catalogue is a versioned transformation with row provenance, flags, crossmatches and missingness.

Radiation, motion and time

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Flux and luminosity** (SFT-ASTRO-FLUX-LUMINOSITY-001): Received flux and source luminosity remain distinct through distance, geometry, band and attenuation records.
- **Astronomical spectrum** (SFT-ASTRO-SPECTRUM-001): A spectrum is an ordered calibrated signal with resolution, response, features and continuum retained.

- **Astronomical redshift** (SFT-ASTRO-REDSHIFT-001): Redshift is an emitted-to-observed feature relation; motion, gravity and expansion causes remain distinguishable.
- **Astrometric and radial motion** (SFT-ASTRO-MOTION-001): Motion retains frame, epochs, baseline, components and uncertainties.
- **Astronomical variability** (SFT-ASTRO-VARIABILITY-001): Variability is source change observed through a declared cadence and window with gaps retained.
- **Periodicity and transient distinction** (SFT-ASTRO-PERIOD-TRANSIENT-001): Periodic recurrence, quasi-periodicity and one-time transience remain distinct temporal classes.

Stellar systems

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Stellar formation** (SFT-ASTRO-STAR-FORMATION-001): Stellar formation is a source-dated assembly history from bound material to sustained stellar state.
- **Stellar structure** (SFT-ASTRO-STELLAR-STRUCTURE-001): A stellar state retains layers, mass, radius, composition, energy transport and observation boundary.
- **Stellar energy generation** (SFT-ASTRO-STELLAR-ENERGY-001): Stellar energy follows species through a complete reaction and loss ledger.
- **Stellar evolution** (SFT-ASTRO-STELLAR-EVOLUTION-001): Stellar evolution is a causal state sequence with initial conditions and mass loss retained.
- **Stellar population** (SFT-ASTRO-STELLAR-POPULATION-001): A stellar population retains selection, distance, extinction, binaries and completeness.
- **Stellar terminal state** (SFT-ASTRO-STELLAR-ENDPOINT-001): Terminal state links progenitor, transition, remnant and ejecta without erasing alternatives.

Planetary systems and small bodies

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Planetary-system identity** (SFT-ASTRO-PLANETARY-SYSTEM-001): A planetary system is a host-and-body organization with orbits, interactions and detection limits retained.
- **Astronomical orbit** (SFT-ASTRO-ORBIT-001): An orbit is a frame- and epoch-bound recurrent state sequence with perturbations and uncertainty.
- **Orbital resonance and tidal state** (SFT-ASTRO-RESONANCE-TIDE-001): Resonance is joint recurrence; tidal locking additionally retains dissipative history.
- **Planet and satellite classification** (SFT-ASTRO-PLANET-001): Planet, satellite and dwarf classifications retain dynamical context and measured features.
- **Exoplanet observation** (SFT-ASTRO-EXOPLANET-001): An exoplanet claim retains host, method, signal, selection, stellar activity and alternatives.
- **Small-body population** (SFT-ASTRO-SMALL-BODY-001): Small-body inference retains survey selection, orbit, activity, composition and completeness.

Interstellar matter, enrichment and formation

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Interstellar-medium identity** (SFT-ASTRO-ISM-001): The interstellar medium is a multiphase carrier with species, dust, fields and transfers retained.

- **Astronomical dust** (SFT-ASTRO-DUST-001): Dust inference retains grain properties, temperature, geometry and radiative method.
- **Molecular cloud** (SFT-ASTRO-MOLECULAR-CLOUD-001): A molecular cloud retains tracer, chemistry, density, motion and boundary.
- **Cosmic chemical enrichment** (SFT-ASTRO-CHEMICAL-ENRICHMENT-001): Enrichment is a source-to-ejecta-to-medium history retaining species, mixing and losses.
- **Environment-to-star-formation relation** (SFT-ASTRO-STAR-FORMATION-ENVIRONMENT-001): Star-formation relations retain region, tracer, time, selection and controls.
- **Stellar feedback** (SFT-ASTRO-FEEDBACK-001): Feedback follows source outputs into medium response with energy, momentum and species ledgers.

Galaxies and populations

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Galaxy identity** (SFT-ASTRO-GALAXY-001): A galaxy is a bound multi-component organization with environment and history retained.
- **Galactic structure** (SFT-ASTRO-GALACTIC-STRUCTURE-001): Galactic structure retains components, projection, kinematics and resolution.
- **Galactic dynamics** (SFT-ASTRO-GALACTIC-DYNAMICS-001): Galactic dynamics retain tracers, baryons, unseen inference, geometry and alternatives.
- **Luminosity-rotation scaling** (SFT-ASTRO-TULLY-FISHER-001): The pre-source rank-four relation is tested against complete source-bound galaxy rows and residuals.
- **Galaxy evolution** (SFT-ASTRO-GALAXY-EVOLUTION-001): Galaxy evolution distinguishes population sequence from individual causal history.
- **Groups, clusters and environment** (SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001): Cluster inference retains membership, gas, potential, selection and mass method.

Compact, high-energy and multimessenger astronomy

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Compact-object identity** (SFT-ASTRO-COMPACT-OBJECT-001): Compact-object identity retains source, environment, signals, method and alternatives.
- **Degenerate remnant classes** (SFT-ASTRO-WHITE-DWARF-NEUTRON-STAR-001): White-dwarf and neutron-star claims retain mass, radius, composition, rotation and field evidence.
- **Black-hole candidate and horizon evidence** (SFT-ASTRO-BLACK-HOLE-001): A black-hole claim retains dynamical mass, compactness, environment, messenger and alternatives.
- **High-energy transient** (SFT-ASTRO-TRANSIENT-001): A transient links source, event, localization, time, bands and afterglow with alternatives.
- **Multi-messenger association** (SFT-ASTRO-MULTIMESSENGER-001): Multi-messenger association retains localization, clocks, selection and background coincidence.
- **Compact-object population** (SFT-ASTRO-COMPACT-POPULATION-001): Compact populations retain detection selection, censoring and source history.

Large-scale structure and expansion

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Large-scale statistical organization** (SFT-ASTRO-COSMIC-HOMOGENEITY-001): Large-scale organization is a survey- and scale-bound tracer distribution.
- **Cosmic-web structure** (SFT-ASTRO-COSMIC-WEB-001): Web structure retains tracer, scale, method, selection and redshift.
- **Cosmic expansion record** (SFT-ASTRO-EXPANSION-RECORD-001): Expansion is a multi-probe distance-redshift record with calibration and covariance retained.
- **Distance-ladder composition** (SFT-ASTRO-DISTANCE-LADDER-001): A distance ladder is an explicit composition of anchors and rungs with covariance.
- **Gravitational-lensing observation** (SFT-ASTRO-LENSING-001): Lensing retains source, lens, observer, geometry, mass model and degeneracies.
- **Expansion-history inference** (SFT-ASTRO-EXPANSION-HISTORY-001): Expansion history is an ordered reconstruction from independently retained probes.

Early universe, background and abundance

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Cosmic thermal history** (SFT-ASTRO-THERMAL-HISTORY-001): Thermal history retains species, interaction, scale, temperature and evidence class by epoch.
- **Cosmic microwave background** (SFT-ASTRO-CMB-001): CMB evidence retains frequency, calibration, beam, masks, foregrounds and covariance.
- **Background acoustic structure** (SFT-ASTRO-ACOUSTIC-STRUCTURE-001): Acoustic structure retains harmonic features, covariance and extraction boundary.
- **Primordial abundance record** (SFT-ASTRO-PRIMORDIAL-ABUNDANCE-001): Primordial abundance inference retains species, sites, methods, depletion and stellar processing.
- **Recombination and photon release** (SFT-ASTRO-RECOMBINATION-001): Recombination links matter state, opacity and photon release with evidence boundaries.
- **Early perturbation record** (SFT-ASTRO-INITIAL-PERTURBATION-001): An early perturbation record retains scale, spectrum, covariance and transfer to later tracers.

Dark sectors, horizons and unobserved claims

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Dark-matter evidence class** (SFT-ASTRO-DARK-MATTER-EVIDENCE-001): Dark-matter evidence is a cross-probe residual with baryons and alternatives retained.
- **Dark-energy evidence class** (SFT-ASTRO-DARK-ENERGY-EVIDENCE-001): Dark-energy evidence retains distance, growth, calibration, covariance and alternatives.
- **Cosmic horizon** (SFT-ASTRO-COSMIC-HORIZON-001): A horizon is an observer- and epoch-bound causal access limit under admitted propagation.
- **Cosmic topology claim** (SFT-ASTRO-TOPOLOGY-001): Topology claims require observable repeated-pattern consequences and controls.
- **Multiverse admissibility boundary** (SFT-ASTRO-MULTIVERSE-001): A multiverse claim enters empirical science only through a distinct observable consequence sealed before data.
- **Cosmic ultimate-fate inference** (SFT-ASTRO-ULTIMATE-FATE-001): Ultimate fate is conditional continuation from current records and admitted laws, not an observed terminal state.

Inference, prediction and disciplinary handoffs

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Universal law and cosmic history distinction** (SFT-ASTRO-LAW-HISTORY-001): Universal law, contingent initial condition and source-dated history remain separate.
- **Astronomical model-observation boundary** (SFT-ASTRO-MODEL-OBSERVATION-001): A model output remains a conditional result compared with, not substituted for, observation.
- **Standing astronomical prediction** (SFT-ASTRO-STANDING-PREDICTION-001): A standing prediction is sealed before outcome and retains every adverse and missing row.
- **Unexplained astronomical observation** (SFT-ASTRO-UNEXPLAINED-001): Unexplained means a source-bound residual after declared controls, not evidence for a preferred unknown cause.
- **Astrobiology and technosignature boundary** (SFT-ASTRO-ASTROBIOLOGY-001): Life or technology claims require replicated signals and complete abiotic, instrumental and contamination controls.
- **Astronomy scientific handoffs** (SFT-ASTRO-ASTRO-HANDOFF-001): Astronomy consumes universal laws and exports source-dated constraints through explicit Physics, Engineering, Earth, Biology and Social boundaries.

Complete claim-by-claim derivations

The following sections are the paper-readable derivation record. The machine packages additionally preserve every one of the 256 candidates and every decision for each claim.

1. Astronomical source

Claim: SFT-ASTRO-SOURCE-001 **Family:** observation_geometry **Statement:** An astronomical source is a bounded carrier whose state produces or alters a registered signal along a declared path.

Forward derivation

Carrier: emitting-or-interacting-source-carrier. Relation: source-state-to-signal-relation.

Retained record: identity-direction-distance-time-and-state-held. Evidence boundary: source-resolution-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

emitting-or-interacting-source-carrier__source-resolution-bounded__source-state-to-signal-relation__identity-direction-distance-time-and-state-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if source identity, direction, distance, time or state is erased.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- CDS-SIMBAD-001 - transport captured; snapshot
sha256: fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.

What it means

This law makes **astronomical source** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 565cccbfaff00ffa6dd9cd475e2289f9cc84f9ac4893d22ee17143533552797e.

Independent certificate:

sha256: 3d8f3e27bb2c7d25363a541d5520a0145eacba97c04b34eac951f364f08cab2b. Engine receipt:

sha256: 5473ed30f43a3ee1099bf327c08e196a9abcd888adb958b68d7c85d51681e958. Candidate count: 256.

Survivor count: 1.

2. Astronomical observer

Claim: SFT-ASTRO-OBSERVER-001 **Family:** `observation_geometry` **Statement:** An observer is a source-located site, platform, instrument and clock relation, not an unlocated view.

Forward derivation

Carrier: `site-platform-instrument-observer-carrier`. Relation: `observer-to-record-relation`.

Retained record: `site-platform-clock-calibration-and-frame-held`. Evidence boundary: `observer-frame-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,

SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SOURCE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

site-platform-instrument-observer-carrier__observer-frame-bounded__observer-to-record-r
 elation__site-platform-clock-calibration-and-frame-held__evidence-class-explicit__root-
 bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if frame, clock, site or calibration changes without record.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
 primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
 sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
 sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- CDS-SIMBAD-001 - transport captured; snapshot
 sha256: fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.

What it means

This law makes **astronomical observer** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:5cb3b2746ad8f0a5c180d9b1e92eadc20c7cf0dc09e86c5cd383007de876bd50.

Independent certificate:

sha256:c85c29c21faa35cbb4d71c03d513d5ffbedd0ec3cfe28874e374d303666c0930. Engine receipt:

sha256:f714bc90e2a60620a46297a6839c7a5b3dbf02d7cd9e51f7b1c81071fb7d0d7a. Candidate count: 256.

Survivor count: 1.

3. Astronomical signal

Claim: SFT-ASTRO-SIGNAL-001 **Family:** observation_geometry **Statement:** A signal retains source, path, band, arrival, detector response and quality status.

Forward derivation

Carrier: source-path-detector-signal-carrier. Relation: propagation-and-detection-relation.

Retained record: emission-path-band-arrival-and-quality-held. Evidence boundary:

signal-channel-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-OBSERVER-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

source-path-detector-signal-carrier__signal-channel-bounded__propagation-and-detection-relation__emission-path-band-arrival-and-quality-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if propagation, detector response or non-detection is omitted.

- false_premise - passed True: reject the generated form lacking the complete physical carrier

- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- CDS-SIMBAD-001 - transport captured; snapshot
sha256: fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.

What it means

This law makes **astronomical signal** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 0f6f0e90e92f6674ca3dfa49c9e355681d6d2a9dc10fae571177321f9787afb9.

Independent certificate:

sha256: 102bcc412ea7545492f737089158b2868a0a554bb5b463b13255d2a6a2bb9623. Engine receipt:

sha256: c0417bce86220840e95fcec8e05c9888b85857274b388151c95c1a744e721b1e. Candidate count: 256.

Survivor count: 1.

4. Angular position and reference frame

Claim: SFT-ASTRO-ANGULAR-POSITION-001 **Family:** `observation_geometry` **Statement:** Angular position is a direction in a named frame and epoch with motion and uncertainty retained.

Forward derivation

Carrier: `direction-coordinate-frame-carrier`. Relation: `frame-transformation-relation`. Retained record: `epoch-origin-basis-motion-and-uncertainty-held`. Evidence boundary: `astrometric-frame-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SIGNAL-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

`direction-coordinate-frame-carrier__astrometric-frame-bounded__frame-transformation-relation__epoch-origin-basis-motion-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule`

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject an unframed coordinate or an epoch-free moving position.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: `fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93`.
- STSCI-MAST-001 - transport captured; snapshot
sha256: `6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701`.
- CDS-SIMBAD-001 - transport captured; snapshot
sha256: `fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f`.

What it means

This law makes **angular position and reference frame** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: `13defca3f4c6a212925722d48d5ce1f4e0de84a75c7fd8824fc566ea36f88edb`.

Independent certificate:

sha256: `ff66a96396bea270a578dbabe975d1f8a2257906b32adf59d807defb4838ad63`. Engine receipt:

sha256: `5baee62478fbc17834102db63297cb4c72efcd2029f90652f9fc24d14588ae8`. Candidate count: 256.

Survivor count: 1.

5. Astronomical distance

Claim: SFT-ASTRO-DISTANCE-001 **Family:** `observation_geometry` **Statement:** Distance is a method-bound source-observer relation whose ladder, calibration and range remain explicit.

Forward derivation

Carrier: source-observer-separation-carrier. Relation: method-specific-distance-inference.
Retained record: method-calibration-ladder-range-and-uncertainty-held. Evidence boundary:
distance-method-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-ANGULAR-POSITION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

source-observer-separation-carrier__distance-method-bounded__method-specific-distance-i
nference__method-calibration-ladder-range-and-uncertainty-held__evidence-class-explicit
__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if one distance method silently substitutes for another.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.

- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- CDS-SIMBAD-001 - transport captured; snapshot
sha256:fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.

What it means

This law makes **astronomical distance** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:20ffa01ddc647a1445bcdaebd7fcedbf5835664239120675f6acef21496031cb.

Independent certificate:

sha256:13fabcb69b206253c29c490a33ebafa64e119c08664615d8b5d19f9e2a17b5e3. Engine receipt:

sha256:163fa182412a88c4e90f0006888a2d78ff4d2fb5d579fe938a5c6e521035aa54. Candidate count: 256.

Survivor count: 1.

6. Survey selection and non-detection

Claim: SFT-ASTRO-SELECTION-001 **Family:** observation_geometry **Statement:** Selection, detection, upper limit, non-detection, missing and outside-footprint status remain distinct.

Forward derivation

Carrier: target-opportunity-detection-status-carrier. Relation: selection-and-detection-relation. Retained record: footprint-cadence-threshold-masks-and-missingness-held. Evidence boundary: survey-selection-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-DISTANCE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

target-opportunity-detection-status-carrier__survey-selection-bounded__selection-and-detection-relation__footprint-cadence-threshold-masks-and-missingness-held__evidence-clas

s-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a non-detection or missing row becomes ordinary zero or confirmed absence.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- CDS-SIMBAD-001 - transport captured; snapshot
sha256: fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.

What it means

This law makes **survey selection and non-detection** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 0ee18d7570070e1263c2b53a4615453b48a38e51d6ddf05fa67de76b802eb06a.

Independent certificate:

sha256: 241fc7a14a377add571071d1a922a384a90dcdd32058c86de97c2f2d2806adec. Engine receipt:

sha256: 2c1a77d0fbc0a3ccd3180f637b9067c4f89fdd1fb2eaf2094bbfd74cf5bcb430. Candidate count: 256.

Survivor count: 1.

7. Astronomical object identity

Claim: SFT-ASTRO-OBJECT-001 **Family:** object_identity_classification **Statement:** Object identity is a reconstructible association of observations across epochs and channels.

Forward derivation

Carrier: cross-epoch-object-carrier. Relation: observation-association-relation. Retained record: positions-times-signals-links-and-alternatives-held. Evidence boundary: association-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,

SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-SELECTION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

cross-epoch-object-carrier__association-bounded__observation-association-relation__positions-times-signals-links-and-alternatives-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if blended or ambiguous detections are forced into one object.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- CDS-SIMBAD-001 - transport captured; snapshot
sha256: fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.
- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical object identity** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The

result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:7933a3cadb9ca1a48e829b5359b5ab4b3682357a96d6e1ad26d017e25e16ec04.

Independent certificate:

sha256:22128bc27aff2fc63ca7224a8596c43bb54f713757b05d3693825f39d09145d6. Engine receipt:

sha256:3aa8c2aa9a6e07c3a509e0f9ddfd46f777bbf74e489bf35ab967a0c7e165d08. Candidate count: 256.

Survivor count: 1.

8. Astronomical classification

Claim: SFT-ASTRO-CLASSIFICATION-001 **Family:** object_identity_classification **Statement:**

Classification retains measured features, method, alternatives and taxonomy version.

Forward derivation

Carrier: feature-evidence-class-carrier. Relation: feature-to-class-relation. Retained record: features-method-probability-alternatives-and-version-held. Evidence boundary: classifier-and-taxonomy-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SELECTION-001, SFT-ASTRO-OBJECT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

feature-evidence-class-carrier__classifier-and-taxonomy-bounded__feature-to-class-relation__features-method-probability-alternatives-and-version-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a label replaces its evidence or an uncertain class becomes certain.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- CDS-SIMBAD-001 - transport captured; snapshot
sha256: fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.
- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical classification** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 47440bf6c83f931457c48db4490812deaac6dce38b46a91f2df5a1b65d62c3cd.

Independent certificate:

sha256: 4001adc5af38a0303fd9786200f61c7e81c9490896370de861253031a95cc7e7. Engine receipt:

sha256: 38676430baf965fd0716b3e5386b3271651d850aff8c29d925e7fe0b9c831e56. Candidate count: 256.

Survivor count: 1.

9. Astronomical population

Claim: SFT-ASTRO-POPULATION-001 **Family:** `object_identity_classification` **Statement:** A population is a selection-bounded set with membership, volume, time, weights and censoring retained.

Forward derivation

Carrier: `selection-bounded-object-set-carrier`. Relation: `membership-and-distribution-relation`. Retained record: `selection-volume-time-weights-and-censoring-held`. Evidence boundary: `population-sampling-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,

SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-SELECTION-001, SFT-ASTRO-CLASSIFICATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

selection-bounded-object-set-carrier__population-sampling-bounded__membership-and-distribution-relation__selection-volume-time-weights-and-censoring-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject population inference from favorable detections alone.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- CDS-SIMBAD-001 - transport captured; snapshot
sha256: fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.
- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical population** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 118634a6115dec00192524edf83a59a07f6bdfdf633f4b36858656624bc0f3395.

Independent certificate:

sha256: bdd657bfd4ff44dc9bbc37bfebda1196b2d7d70fd21298cb2620d26fb6a6e3c0. Engine receipt:

sha256: 0a62514b2e54766ac4b30eedccd4c6ce170a42980849b27b1ddda55d3fa31441. Candidate count: 256.

Survivor count: 1.

10. Astronomical morphology

Claim: SFT-ASTRO-MORPHOLOGY-001 **Family:** object_identity_classification **Statement:** Morphology is a resolution- and band-bound organization of distinguishable components.

Forward derivation

Carrier: resolved-structure-carrier. Relation: component-adjacency-shape-relation. Retained record: resolution-band-orientation-distance-and-uncertainty-held. Evidence boundary: morphology-method-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SELECTION-001, SFT-ASTRO-POPULATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

resolved-structure-carrier__morphology-method-bounded__component-adjacency-shape-relation__resolution-band-orientation-distance-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if resolution or projection is erased.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The

post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- CDS-SIMBAD-001 - transport captured; snapshot
sha256: fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.
- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical morphology** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: e88fe6a6c77d63f83bac0099b5092fb4a1438c2f09238322df9cd1049e2007d8.

Independent certificate:

sha256: 4d77c06617da2c0ea2a9bd50df76468bc2f21ccf8919459a7309beae221ade1b. Engine receipt:

sha256: 7e5ce0ee426488dc2e6d78aa5c6d1fca06d6cdccccfa3f0bd56b3eafa0fe44dc. Candidate count: 256.

Survivor count: 1.

11. Astronomical composition inference

Claim: SFT-ASTRO-COMPOSITION-001 **Family:** object_identity_classification **Statement:** Composition inference retains species evidence, excitation, method and alternatives.

Forward derivation

Carrier: species-feature-source-carrier. Relation: spectral-or-particle-composition-relation.

Retained record: species-lines-model-boundary-abundance-and-errors-held. Evidence boundary: composition-method-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-SELECTION-001, SFT-ASTRO-MORPHOLOGY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

species-feature-source-carrier__composition-method-bounded__spectral-or-particle-composition-relation__species-lines-model-boundary-abundance-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a line identification or abundance is detached from its method.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- CDS-SIMBAD-001 - transport captured; snapshot
sha256: fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.
- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical composition inference** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 97b806c8c263910ff4c3d0ad494c1381d9981221b5a79b9cd72c0f5f2e16b4c6.

Independent certificate:

sha256: 4878bae7879c087c530b2f96c7487e1e7c9e1132faf6c55191b76d20c3d4a551. Engine receipt:

sha256: 26cc87a10853fecab01b32b1fa0e693cfbf86fd0cbec19091ee16fc2fdf5d17d. Candidate count: 256.

Survivor count: 1.

12. Astronomical catalogue

Claim: SFT-ASTRO-CATALOG-001 **Family:** object_identity_classification **Statement:** A catalogue is a versioned transformation with row provenance, flags, crossmatches and missingness.

Forward derivation

Carrier: versioned-row-provenance-carrier. Relation: source-to-catalogue-transformation. Retained record: query-version-columns-flags-crossmatches-and-missingness-held. Evidence boundary: catalogue-release-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
 SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
 SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
 SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-SELECTION-001, SFT-ASTRO-COMPOSITION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

versioned-row-provenance-carrier__catalogue-release-bounded__source-to-catalogue-transf
 ormation__query-version-columns-flags-crossmatches-and-missingness-held__evidence-class
 -explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-e
 xtra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if catalogue version, query or quality filtering is hidden.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
 primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- CDS-SIMBAD-001 - transport captured; snapshot
 sha256: fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.
- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
 sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.

- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical catalogue** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:97174d95ba56d68090edccc04e6907baf0593c34bb3fe9fcd1b466964055235e.

Independent certificate:

sha256:0c92420b66a9497d20c6afdd1cc10dad772cfdb7aeec9d19fc8595a697035400. Engine receipt:

sha256:acdd861f1b9ddfb158859329b8062e970f19f77f497b673fe8caf0c4272f68bc. Candidate count: 256.

Survivor count: 1.

13. Flux and luminosity

Claim: SFT-ASTRO-FLUX-LUMINOSITY-001 **Family:** radiation_motion_time **Statement:** Received flux and source luminosity remain distinct through distance, geometry, band and attenuation records.

Forward derivation

Carrier: source-path-received-radiation-carrier. Relation: flux-distance-luminosity-relation.

Retained record: band-distance-geometry-extinction-and-time-held. Evidence boundary: radiometric-boundary-declared.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-CATALOG-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

source-path-received-radiation-carrier__radiometric-boundary-declared__flux-distance-luminosity-relation__band-distance-geometry-extinction-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if apparent brightness is relabelled intrinsic luminosity.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **flux and luminosity** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: b19c3ed32c80f424415e8673ff8dda6605c3c09393a46c8f3c53898f8cd9a1dd.

Independent certificate:

sha256: 1b7b2f2236e45abf9ccd59bf7988e7e485417753a0485bb211fba17e06e513f1. Engine receipt:

sha256: 244626352d46e53db6f803dce4d8de4b5a9fc01a0c4a48c9ee7c74863b9f3f0a. Candidate count: 256.

Survivor count: 1.

14. Astronomical spectrum

Claim: SFT-ASTRO-SPECTRUM-001 **Family:** `radiation_motion_time` **Statement:** A spectrum is an ordered calibrated signal with resolution, response, features and continuum retained.

Forward derivation

Carrier: `ordered-band-signal-carrier`. Relation: `wavelength-frequency-energy-feature-relation`.

Retained record: `resolution-response-calibration-lines-and-continuum-held`. Evidence boundary: `spectral-method-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,

SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-CATALOG-001, SFT-ASTRO-FLUX-LUMINOSITY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

ordered-band-signal-carrier__spectral-method-bounded__wavelength-frequency-energy-feature-relation__resolution-response-calibration-lines-and-continuum-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject an uncalibrated or selectively cropped feature set.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical spectrum** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result

may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:86ba6eeb0195e117e0f21b349976ca5a1666e8e5f1b2efcbd0150b49dcb56aa0.

Independent certificate:

sha256:a15145a02e581c0956ac90ef940158e7379a804df468c502edbd530448dc7011. Engine receipt:

sha256:3c2ae757898fd4ca1063639281403d361117113843e1a4bd69144ffa55655594. Candidate count: 256.

Survivor count: 1.

15. Astronomical redshift

Claim: SFT-ASTRO-REDSHIFT-001 **Family:** radiation_motion_time **Statement:** Redshift is an emitted-to-observed feature relation; motion, gravity and expansion causes remain distinguishable.

Forward derivation

Carrier: emitted-observed-feature-carrier. Relation: feature-displacement-relation. Retained record: feature-identity-frame-motion-gravity-expansion-and-error-held. Evidence boundary: redshift-cause-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-CATALOG-001, SFT-ASTRO-SPECTRUM-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

emitted-observed-feature-carrier__redshift-cause-bounded__feature-displacement-relation
__feature-identity-frame-motion-gravity-expansion-and-error-held__evidence-class-explic
it__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-ru
le

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if measured redshift alone is treated as a unique distance or cause.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: `false`.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical redshift** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: ed2a05b04907340d5a36441d36f3e6ea31ff63ed5689d61e0e2012dd605ca85f.

Independent certificate:

sha256: 35bfc901673ec988fb19ffa949a00854f8c17c3ad8c992741939f62ad35a845e. Engine receipt:

sha256: d3148d125f87843df676f13fcd10f8c9c05baa3afe48398d6743fa616d5606c2. Candidate count: 256.

Survivor count: 1.

16. Astrometric and radial motion

Claim: SFT-ASTRO-MOTION-001 **Family:** `radiation_motion_time` **Statement:** Motion retains frame, epochs, baseline, components and uncertainties.

Forward derivation

Carrier: `time-indexed-position-spectrum-carrier`. Relation: `position-or-feature-change-relation`. Retained record: `frame-epochs-baseline-components-and-errors-held`. Evidence boundary: `motion-observation-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,

SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CATALOG-001, SFT-ASTRO-REDSHIFT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

time-indexed-position-spectrum-carrier__motion-observation-bounded__position-or-feature-change-relation__frame-epochs-baseline-components-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if apparent and physical motion are conflated.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astrometric and radial motion** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: d3a960c27dc6800e5495d8cdee40bae9b4e9435e5a02a8fd0063472a90888a38.

Independent certificate:

sha256: 0ba8e7ef3cbe87c6192cf84b713fdc99de580170846a34cda190a5816a8c8080. Engine receipt:

sha256: bbea1c97377efd636dc90e8df5d9963b2b6f55a59422efc302d0a0172897de8f. Candidate count: 256.

Survivor count: 1.

17. Astronomical variability

Claim: SFT-ASTRO-VARIABILITY-001 **Family:** radiation_motion_time **Statement:** Variability is source change observed through a declared cadence and window with gaps retained.

Forward derivation

Carrier: time-series-source-carrier. Relation: state-to-signal-change-relation. Retained record: cadence-window-band-errors-and-nondetections-held. Evidence boundary: time-series-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CATALOG-001, SFT-ASTRO-MOTION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

time-series-source-carrier__time-series-bounded__state-to-signal-change-relation__cadence-window-band-errors-and-nondetections-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if sampling aliases or missing intervals are erased.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical variability** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: f73f34c8d8bfaedff13e3ac0a5600eeddd28072872610c552df5fadfc94a40be.

Independent certificate:

sha256: 310ff1e5bdfbf1862e8f7b9b9f56470f6b58f09944fd7b5fa7700f8c4eacd9c4. Engine receipt:

sha256: 91d8d1c51df7727d0da50e2ebc76099744b8faca1c4dafd29fc99b962fc72aeb. Candidate count: 256.

Survivor count: 1.

18. Periodicity and transient distinction

Claim: SFT-ASTRO-PERIOD-TRANSIENT-001 **Family:** radiation_motion_time **Statement:** Periodic recurrence, quasi-periodicity and one-time transience remain distinct temporal classes.

Forward derivation

Carrier: time-ordered-event-or-cycle-carrier. Relation: recurrence-or-one-event-relation. Retained record: epochs-phase-window-baseline-and-alternatives-held. Evidence boundary: temporal-class-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-CATALOG-001, SFT-ASTRO-VARIABILITY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

time-ordered-event-or-cycle-carrier__temporal-class-bounded__recurrence-or-one-event-relation__epochs-phase-window-baseline-and-alternatives-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a finite window manufactures a period or erases recurrence.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **periodicity and transient distinction** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: a15f56a691d4023dd5e50cf4fbedaadff6cc460c687b3ad771e8775e1d2af1bf.

Independent certificate:

sha256: c2b118aa01b580633cb3940fd3df7efea5fbd532e4c687cf6317519de49e28d3. Engine receipt:

sha256: d4a7ebf3b590fe9908e73b46d032cbcb0e34ef99ad286ab3ff5644df05ac1b6a. Candidate count: 256.

Survivor count: 1.

19. Stellar formation

Claim: SFT-ASTRO-STAR-FORMATION-001 **Family:** stellar_systems **Statement:** Stellar formation is a source-dated assembly history from bound material to sustained stellar state.

Forward derivation

Carrier: bound-collapsing-gas-dust-carrier. Relation:

assembly-heating-cooling-ignition-relation. Retained record:

environment-mass-flow-time-and-evidence-held. Evidence boundary: formation-history-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,

SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,

SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
 SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-PERIOD-TRANSIENT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

bound-collapsing-gas-dust-carrier__formation-history-bounded__assembly-heating-cooling-
 ignition-relation__environment-mass-flow-time-and-evidence-held__evidence-class-explici
 t__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rul
 e

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a final star is used as its own formation record.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
 primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
 sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
 sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-HEASARC-001 - transport captured; snapshot
 sha256: 030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar formation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:aff9065f6aff056a81855cc6cc00c0471ec33aaaed49b5bc052237fd599cb571.

Independent certificate:

sha256:e762c8672780047e937e74efff63c138ea6ef388f194592a653480c9407d49b4. Engine receipt:

sha256:1687511f165df988b1a70cb12f64fda63bc0a8236daab845dd92d2f7a1723756. Candidate count: 256.

Survivor count: 1.

20. Stellar structure

Claim: SFT-ASTRO-STELLAR-STRUCTURE-001 **Family:** stellar_systems **Statement:** A stellar state retains layers, mass, radius, composition, energy transport and observation boundary.

Forward derivation

Carrier: layered-stellar-state-carrier. Relation: gravity-pressure-energy-transport-balance.

Retained record: mass-radius-composition-temperature-luminosity-and-age-held. Evidence boundary: stellar-model-observation-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-PERIOD-TRANSIENT-001, SFT-ASTRO-STAR-FORMATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

layered-stellar-state-carrier__stellar-model-observation-bounded__gravity-pressure-energy-transport-balance__mass-radius-composition-temperature-luminosity-and-age-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if model state is relabelled direct interior observation.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-HEASARC-001 - transport captured; snapshot
sha256: 030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar structure** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: a3997f5afb8a2b01da60edf201dc2f5140aeaa2718c3ad44c3522a8e3390119d.

Independent certificate:

sha256: 54e7a20b55313ee7e0e5a15ab6b668abab7fd672ebcc20c89779e2b064f734ea. Engine receipt:

sha256: 745723e85c03326d670183c817d64823618a9576ecd1efb2ace244d954efa0b5. Candidate count: 256.

Survivor count: 1.

21. Stellar energy generation

Claim: SFT-ASTRO-STELLAR-ENERGY-001 **Family:** stellar_systems **Statement:** Stellar energy follows species through a complete reaction and loss ledger.

Forward derivation

Carrier: fuel-reaction-product-radiation-carrier. Relation: reaction-network-to-output-ledger. Retained record: species-rates-losses-time-and-neutrino-photon-record-held. Evidence boundary: nuclear-network-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,

SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-PERIOD-TRANSIENT-001, SFT-ASTRO-STELLAR-STRUCTURE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

`fuel-reaction-product-radiation-carrier__nuclear-network-bounded__reaction-network-to-output-ledger__species-rates-losses-time-and-neutrino-photon-record-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule`

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if missing energy or species is fitted away.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
`sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.`
- STSCI-MAST-001 - transport captured; snapshot
`sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.`
- NASA-HEASARC-001 - transport captured; snapshot
`sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.`

What it means

This law makes **stellar energy generation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: `sha256:af66d66442de0b714d0391fe86b6318428662926d24429e99f8fb3b8b545e863.`

Independent certificate:

`sha256:94ada775f758ba03641cf228ce143545c1981c0d944f1fb960e597d1bbd44656.` Engine receipt:

sha256:677612669b5fd145ef82ad8bd915e8ef433e4a8c491ffc1b3467467d9f4db940. Candidate count: 256.
Survivor count: 1.

22. Stellar evolution

Claim: SFT-ASTRO-STELLAR-EVOLUTION-001 **Family:** stellar_systems **Statement:** Stellar evolution is a causal state sequence with initial conditions and mass loss retained.

Forward derivation

Carrier: time-ordered-stellar-state-carrier. Relation: fuel-structure-transition-relation.
Retained record: initial-mass-composition-environment-loss-and-time-held. Evidence boundary: evolution-track-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-PERIOD-TRANSIENT-001, SFT-ASTRO-STELLAR-ENERGY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

time-ordered-stellar-state-carrier__evolution-track-bounded__fuel-structure-transition-relation__initial-mass-composition-environment-loss-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a population ordering is silently treated as one star's history.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-HEASARC-001 - transport captured; snapshot
sha256: 030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar evolution** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 66c108b2959a933578e09e963e9f54983c00118989c9a3768ebbbf6cc6b9a3d1.

Independent certificate:

sha256: 02257c3ab768120eb20d12b1c5efa8c2a6869f8e4f235dc42ae8faa9ec0c6b3b. Engine receipt:

sha256: 452f1a61686abff87da13d1345308ad9722fd3c30e2db5ed71e63dd359a48f42. Candidate count: 256.

Survivor count: 1.

23. Stellar population

Claim: SFT-ASTRO-STELLAR-POPULATION-001 **Family:** `stellar_systems` **Statement:** A stellar population retains selection, distance, extinction, binaries and completeness.

Forward derivation

Carrier: `selection-bounded-star-set-carrier`. Relation:

`age-metallicity-mass-distribution-relation`. Retained record:

`selection-distance-extinction-binaries-and-completeness-held`. Evidence boundary:

`population-synthesis-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-PERIOD-TRANSIENT-001, SFT-ASTRO-STELLAR-EVOLUTION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the

registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

selection-bounded-star-set-carrier__population-synthesis-bounded__age-metallicity-mass-distribution-relation__selection-distance-extinction-binaries-and-completeness-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if unresolved binaries or selection effects are ignored.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-HEASARC-001 - transport captured; snapshot
sha256: 030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar population** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 048d32b7168b1f7fffd3213a961342a616910b7c1dc1d3a69081b2234eec2ad2.

Independent certificate:

sha256: e63ea43bb6eaa3b1a3117653309d670e5702818439949fb0d25311758ea2d48d. Engine receipt:

sha256: 238a4d5f7c1962eea3c0039c054810fe25aa55a0f57218b3ac96885f99bce77a. Candidate count: 256.

Survivor count: 1.

24. Stellar terminal state

Claim: SFT-ASTRO-STELLAR-ENDPOINT-001 **Family:** stellar_systems **Statement:** Terminal state links progenitor, transition, remnant and ejecta without erasing alternatives.

Forward derivation

Carrier: progenitor-transition-remnant-carrier. Relation: mass-composition-loss-collapse-relation. Retained record: progenitor-event-remnant-ejecta-and-uncertainty-held. Evidence boundary: endpoint-class-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-PERIOD-TRANSIENT-001, SFT-ASTRO-STELLAR-POPULATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

progenitor-transition-remnant-carrier__endpoint-class-bounded__mass-composition-loss-collapse-relation__progenitor-event-remnant-ejecta-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if remnant class is inferred without progenitor and event boundaries.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-HEASARC-001 - transport captured; snapshot
sha256: 030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar terminal state** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 9fe62de74937a175eebe6025cf41ad0ab9ba8ec4489138c425a2961fbefcfc98.

Independent certificate:

sha256: 77800a391e1523af6300e79dec6f568d73005b769f6cb9852d1660fdd03babf9. Engine receipt:

sha256: cc3186110029ec8264d7cfff7583e93cdf8cb599853d2645374bc9c7377ad2f8. Candidate count: 256.

Survivor count: 1.

25. Planetary-system identity

Claim: SFT-ASTRO-PLANETARY-SYSTEM-001 **Family:** planetary_small_body **Statement:** A planetary system is a host-and-body organization with orbits, interactions and detection limits retained.

Forward derivation

Carrier: host-orbiting-body-joint-carrier. Relation: bound-orbit-and-interaction-relation.

Retained record: host-bodies-orbits-masses-epochs-and-errors-held. Evidence boundary: system-detection-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-STELLAR-ENDPOINT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

host-orbiting-body-joint-carrier__system-detection-bounded__bound-orbit-and-interaction-relation__host-bodies-orbits-masses-epochs-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if candidates or unseen completeness are treated as confirmed bodies.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-EXOPLANET-ARCHIVE-001 - transport captured; snapshot
sha256: 78cd771cc5a4ed3f31f381048207300e09c5832c919181d49af67c0ac1e83d8c.
- MPC-DATA-SERVICES-001 - transport captured; snapshot
sha256: 107d2925b2ff051209d57c22181d1b485fd7f09b1ced9c5c79a49542668e2d77.
- NASA-JPL-HORIZONS-001 - transport captured; snapshot
sha256: ae7b2f15acd48d137f7e261e8eef9a44faf6c72a655a28aeae5ee9a0a8bb928c.

What it means

This law makes **planetary-system identity** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: f36b76be1077d29fc769d1f1ab00606d6c124d775083225954c54b0ceccea027.

Independent certificate:

sha256: 434327a0a8c1fb0f7eef01faa80e7fedbf3ba30a9807b0548fd323e0331ad6ca. Engine receipt:

sha256: c246e3a9b32c67d57ac4ec24cf388171bc7f56b403018ce48d1834def18b7d70. Candidate count: 256.

Survivor count: 1.

26. Astronomical orbit

Claim: SFT-ASTRO-ORBIT-001 **Family:** planetary_small_body **Statement:** An orbit is a frame- and epoch-bound recurrent state sequence with perturbations and uncertainty.

Forward derivation

Carrier: body-primary-state-sequence-carrier. Relation: bounded-recurrent-motion-relation.

Retained record: frame-epoch-state-elements-perturbations-and-errors-held. Evidence boundary: orbit-solution-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,

SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
 SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-STELLAR-ENDPOINT-001, SFT-ASTRO-PLANETARY-SYSTEM-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

body-primary-state-sequence-carrier__orbit-solution-bounded__bounded-recurrent-motion-relation__frame-epoch-state-elements-perturbations-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if fitted elements replace observations or an open trajectory is called bound.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-EXOPLANET-ARCHIVE-001 - transport captured; snapshot
sha256: 78cd771cc5a4ed3f31f381048207300e09c5832c919181d49af67c0ac1e83d8c.
- MPC-DATA-SERVICES-001 - transport captured; snapshot
sha256: 107d2925b2ff051209d57c22181d1b485fd7f09b1ced9c5c79a49542668e2d77.
- NASA-JPL-HORIZONS-001 - transport captured; snapshot
sha256: ae7b2f15acd48d137f7e261e8eef9a44faf6c72a655a28aeae5ee9a0a8bb928c.

What it means

This law makes **astronomical orbit** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:7b62450e2c54408cebec61a6742b4589ffe9255d89d29b92612b494064f5dad6.

Independent certificate:

sha256:7532424071c5951225ee7eeda4a039643700bd7619ec2815e03d65eaa3a4e2c. Engine receipt:

sha256:058bed515488bd9ed3e3f8e3a7f6cdbe218fef342910a86800cf6f40202f4be1. Candidate count: 256.

Survivor count: 1.

27. Orbital resonance and tidal state

Claim: SFT-ASTRO-RESONANCE-TIDE-001 **Family:** planetary_small_body **Statement:** Resonance is joint recurrence; tidal locking additionally retains dissipative history.

Forward derivation

Carrier: coupled-orbit-spin-carrier. Relation: integer-recurrence-and-dissipation-relation.

Retained record: periods-phases-bodies-history-and-errors-held. Evidence boundary: resonance-baseline-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-STELLAR-ENDPOINT-001, SFT-ASTRO-ORBIT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

coupled-orbit-spin-carrier__resonance-baseline-bounded__integer-recurrence-and-dissipation-relation__periods-phases-bodies-history-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject a near ratio without phase recurrence or a lock without history.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-EXOPLANET-ARCHIVE-001 - transport captured; snapshot
sha256: 78cd771cc5a4ed3f31f381048207300e09c5832c919181d49af67c0ac1e83d8c.
- MPC-DATA-SERVICES-001 - transport captured; snapshot
sha256: 107d2925b2ff051209d57c22181d1b485fd7f09b1ced9c5c79a49542668e2d77.
- NASA-JPL-HORIZONS-001 - transport captured; snapshot
sha256: ae7b2f15acd48d137f7e261e8eef9a44faf6c72a655a28aeae5ee9a0a8bb928c.

What it means

This law makes **orbital resonance and tidal state** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: a2c94e58be574845e88fa39a7bda07911bb4d45ec04953094315bfeed6f149ed.

Independent certificate:

sha256: 83ad41de52b1a012f02ae2c6785ab09d5f4562e655af758c740080a9bbac8afe. Engine receipt:

sha256: 7fb8371722eb7ced1d07e8e9617f2a368b04dd05adda017db98648d339afa87b. Candidate count: 256.

Survivor count: 1.

28. Planet and satellite classification

Claim: SFT-ASTRO-PLANET-001 **Family:** `planetary_small_body` **Statement:** Planet, satellite and dwarf classifications retain dynamical context and measured features.

Forward derivation

Carrier: `body-orbit-mass-shape-context-carrier`. Relation: `feature-and-dynamical-class-relation`. Retained record: `host-orbit-mass-radius-composition-and-method-held`. Evidence boundary: `classification-version-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,

SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-STELLAR-ENDPOINT-001, SFT-ASTRO-RESONANCE-TIDE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

body-orbit-mass-shape-context-carrier__classification-version-bounded__feature-and-dyna
 mical-class-relation__host-orbit-mass-radius-composition-and-method-held__evidence-clas
 s-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-
 extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if taxonomy supplies a physical law or uncertain mass is hidden.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
 primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The
 post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-EXOPLANET-ARCHIVE-001 - transport captured; snapshot
 sha256: 78cd771cc5a4ed3f31f381048207300e09c5832c919181d49af67c0ac1e83d8c.
- MPC-DATA-SERVICES-001 - transport captured; snapshot
 sha256: 107d2925b2ff051209d57c22181d1b485fd7f09b1ced9c5c79a49542668e2d77.
- NASA-JPL-HORIZONS-001 - transport captured; snapshot
 sha256: ae7b2f15acd48d137f7e261e8eef9a44faf6c72a655a28aeae5ee9a0a8bb928c.

What it means

This law makes **planet and satellite classification** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 6ca7cc9c45829a03f3f63a9b4f14c32d0d9af2734b2567346b248ba7f882853c.

Independent certificate:

sha256: 179b1e4acc8322590db12527333eb316be24380622582a423afc39ef77b984e1. Engine receipt:

sha256:cc44ff010d51d2f18e183a9642d1c1f1bc3525ea658bd18e92973b259c69c90e. Candidate count: 256.
Survivor count: 1.

29. Exoplanet observation

Claim: SFT-ASTRO-EXOPLANET-001 **Family:** planetary_small_body **Statement:** An exoplanet claim retains host, method, signal, selection, stellar activity and alternatives.

Forward derivation

Carrier: host-signal-planet-candidate-carrier. Relation:
transit-radial-imaging-or-timing-inference. Retained record:
method-selection-host-activity-period-and-errors-held. Evidence boundary:
detection-method-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-STELLAR-ENDPOINT-001, SFT-ASTRO-PLANET-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

host-signal-planet-candidate-carrier__detection-method-bounded__transit-radial-imaging-or-timing-inference__method-selection-host-activity-period-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if an instrumental or stellar signal is silently promoted to a planet.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor

- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-EXOPLANET-ARCHIVE-001 - transport captured; snapshot
sha256: 78cd771cc5a4ed3f31f381048207300e09c5832c919181d49af67c0ac1e83d8c.
- MPC-DATA-SERVICES-001 - transport captured; snapshot
sha256: 107d2925b2ff051209d57c22181d1b485fd7f09b1ced9c5c79a49542668e2d77.
- NASA-JPL-HORIZONS-001 - transport captured; snapshot
sha256: ae7b2f15acd48d137f7e261e8eef9a44faf6c72a655a28aeae5ee9a0a8bb928c.

What it means

This law makes **exoplanet observation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: ea5ead6a98089883080166720f37ddb856b12ac8fe7c3ff5266465668054f8e6.

Independent certificate:

sha256: 843caea42accb3576284810f3fbdeb7d2240a1dbcabd8601ab3d3dff2af759a1. Engine receipt:

sha256: b45e633c526545940df83343788a26f671c0061ddd899a9fae70251d37564314. Candidate count: 256.

Survivor count: 1.

30. Small-body population

Claim: SFT-ASTRO-SMALL-BODY-001 **Family:** planetary_small_body **Statement:** Small-body inference retains survey selection, orbit, activity, composition and completeness.

Forward derivation

Carrier: selection-bounded-comet-asteroid-carrier. Relation:
orbit-size-composition-population-relation. Retained record:
survey-selection-orbits-activity-and-completeness-held. Evidence boundary:
small-body-survey-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-STELLAR-ENDPOINT-001, SFT-ASTRO-EXOPLANET-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

selection-bounded-comet-asteroid-carrier__small-body-survey-bounded__orbit-size-composition-population-relation__survey-selection-orbits-activity-and-completeness-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if detected objects are treated as the complete population.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-EXOPLANET-ARCHIVE-001 - transport captured; snapshot
sha256: 78cd771cc5a4ed3f31f381048207300e09c5832c919181d49af67c0ac1e83d8c.
- MPC-DATA-SERVICES-001 - transport captured; snapshot
sha256: 107d2925b2ff051209d57c22181d1b485fd7f09b1ced9c5c79a49542668e2d77.
- NASA-JPL-HORIZONS-001 - transport captured; snapshot
sha256: ae7b2f15acd48d137f7e261e8eef9a44faf6c72a655a28aeae5ee9a0a8bb928c.

What it means

This law makes **small-body population** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 19cae08f9d417aacc353afa84f0880d9d08950e3d1d61efd9dc8c13b999c86f.

Independent certificate:

sha256: 4630616a953f618776cbb45fce376333cd1132502480628ac2790624ff60ac6b. Engine receipt:

sha256: dc8a555a924c996226270750247dc628b5b4d7741b40acd27df07fd8015d85b9. Candidate count: 256.

Survivor count: 1.

31. Interstellar-medium identity

Claim: SFT-ASTRO-ISM-001 **Family:** interstellar_enrichment_formation **Statement:** The interstellar medium is a multiphase carrier with species, dust, fields and transfers retained.

Forward derivation

Carrier: multiphase-gas-dust-field-carrier. Relation: phase-and-transfer-network. Retained record: location-phase-species-field-scale-and-time-held. Evidence boundary: ism-observation-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SMALL-BODY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

multiphase-gas-dust-field-carrier__ism-observation-bounded__phase-and-transfer-network__location-phase-species-field-scale-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if phases or unseen components are collapsed into one medium.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **interstellar-medium identity** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:2e0084bbdf6cea58e5bc389d076e39f953de8804dba7bcc741f8882daa105b3d.

Independent certificate:

sha256:460f0407f41416cdce4b9f9832e47c32d6534f0350199301a6d1711d2b8ccad7. Engine receipt:

sha256:0cec1bfffce82ee255eab56babbd859767b0618e7791c555b77d4a3cc8a0467e4. Candidate count: 256.

Survivor count: 1.

32. Astronomical dust

Claim: SFT-ASTRO-DUST-001 **Family:** interstellar_enrichment_formation **Statement:** Dust inference retains grain properties, temperature, geometry and radiative method.

Forward derivation

Carrier: grain-composition-size-temperature-carrier. Relation:

absorption-emission-scattering-relation. Retained record:

composition-size-band-temperature-and-geometry-held. Evidence boundary: dust-model-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-SMALL-BODY-001, SFT-ASTRO-ISM-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

grain-composition-size-temperature-carrier__dust-model-bounded__absorption-emission-scattering-relation__composition-size-band-temperature-and-geometry-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if extinction and emission models are conflated.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **astronomical dust** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:0c2e129fcb962f63506b440574274f57b43cede8b208645d3212445dbb7a6868.

Independent certificate:

sha256:3a98bd8a09d93ebecfbe23948092e24b3ad1a07b107a338132cf3f43b6c739fe. Engine receipt:

sha256:213ab20c5ab76aa3761d091100564d21726e2112b6030f48c4d30a319b19f67d. Candidate count: 256.

Survivor count: 1.

33. Molecular cloud

Claim: SFT-ASTRO-MOLECULAR-CLOUD-001 **Family:** `interstellar_enrichment_formation` **Statement:** A molecular cloud retains tracer, chemistry, density, motion and boundary.

Forward derivation

Carrier: `molecule-gas-dust-bound-region-carrier`. Relation: `chemistry-cooling-collapse-relation`.

Retained record: `species-density-temperature-motion-and-boundary-held`. Evidence boundary: `tracer-method-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
 SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
 SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
 SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-SMALL-BODY-001, SFT-ASTRO-DUST-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

molecule-gas-dust-bound-region-carrier__tracer-method-bounded__chemistry-cooling-collapse-relation__species-density-temperature-motion-and-boundary-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if one tracer is assumed to equal total material without a bridge.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- STSCI-MAST-001 - transport captured; snapshot
 sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-IRSA-001 - transport captured; snapshot
 sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.

- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **molecular cloud** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:36286860fb3134180dfeae0f491055251b94c86c268c6ac01a3beecef5efd7bb7.

Independent certificate:

sha256:76689cd892a0727918bd008bd57690e38ba65658cb07489e733e72b5059ed00a. Engine receipt:

sha256:05313d27771f58964ebe43b714f0f74af495152b0ad8fc9362c3a0d5e9376fae. Candidate count: 256.

Survivor count: 1.

34. Cosmic chemical enrichment

Claim: SFT-ASTRO-CHEMICAL-ENRICHMENT-001 **Family:** interstellar_enrichment_formation **Statement:** Enrichment is a source-to-ejecta-to-medium history retaining species, mixing and losses.

Forward derivation

Carrier: source-ejecta-medium-generation-carrier. Relation: nucleosynthesis-transfer-mixing-relation. Retained record: species-yields-sources-time-mixing-and-losses-held. Evidence boundary: enrichment-history-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SMALL-BODY-001, SFT-ASTRO-MOLECULAR-CLOUD-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

source-ejecta-medium-generation-carrier__enrichment-history-bounded__nucleosynthesis-transfer-mixing-relation__species-yields-sources-time-mixing-and-losses-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if abundance similarity proves unique provenance.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-IRSA-001 - transport captured; snapshot
sha256: 1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.
- NASA-HEASARC-001 - transport captured; snapshot
sha256: 030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **cosmic chemical enrichment** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: c7db598e64b5fbc25cf346955a54a6f0e77bc79cd86c9aea3d4c0aa9cf6a55de.

Independent certificate:

sha256: 96a7b74e90ec9eb47e5498a77dcdb07a931b2ba670704ebfecde71922425fc63. Engine receipt:

sha256: 9cd403fe963694ee2817abb48d9e1c87cb7b38b49e61eb90c21eff8726bc85be. Candidate count: 256.

Survivor count: 1.

35. Environment-to-star-formation relation

Claim: SFT-ASTRO-STAR-FORMATION-ENVIRONMENT-001 **Family:** `interstellar_enrichment_formation`

Statement: Star-formation relations retain region, tracer, time, selection and controls.

Forward derivation

Carrier: `cloud-environment-young-star-carrier`. Relation: `condition-to-formation-rate-comparison`.

Retained record: `mass-area-time-tracer-selection-and-controls-held`. Evidence boundary:

`region-and-tracer-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,

SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,

SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,

SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,

SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,

SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-SMALL-BODY-001, SFT-ASTRO-CHEMICAL-ENRICHMENT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

cloud-environment-young-star-carrier__region-and-tracer-bounded__condition-to-formation
 -rate-comparison__mass-area-time-tracer-selection-and-controls-held__evidence-class-exp
 licit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra
 -rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject a universal rate from one tracer or region.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
 primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- STSCI-MAST-001 - transport captured; snapshot
 sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-IRSA-001 - transport captured; snapshot
 sha256: 1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.
- NASA-HEASARC-001 - transport captured; snapshot
 sha256: 030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **environment-to-star-formation relation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific

chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:7c1212c0046c03473acf1b29db6ed405fa824c9392d36ddb63ff27f172a1086a.

Independent certificate:

sha256:96e562baf26a6e7ede30895e728187ccf22a0694adf5f66aa414052b302bf80b. Engine receipt:

sha256:28da28a0062a3873e96d08d96610ca8b8af37164f59251da7046dc4a27df6293. Candidate count: 256.

Survivor count: 1.

36. Stellar feedback

Claim: SFT-ASTRO-FEEDBACK-001 **Family:** interstellar_enrichment_formation **Statement:** Feedback follows source outputs into medium response with energy, momentum and species ledgers.

Forward derivation

Carrier: star-output-medium-response-carrier. Relation: radiation-wind-ejecta-transfer-relation. Retained record: source-energy-momentum-species-geometry-and-time-held. Evidence boundary: feedback-domain-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SMALL-BODY-001, SFT-ASTRO-STAR-FORMATION-ENVIRONMENT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

star-output-medium-response-carrier__feedback-domain-bounded__radiation-wind-ejecta-transfer-relation__source-energy-momentum-species-geometry-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if correlation alone establishes direction or conserved transfer.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: `false`.

- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar feedback** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:8dfee6a28b68fc8947dce890445c7cceb14985d5ddffc3e5c510f6c9aea09bb7.

Independent certificate:

sha256:d3a6a3293cd9fb267f7a6c25398812d1229b640d12be01f724fd727c827fad30. Engine receipt:

sha256:1a44a2f07fadd9785f8d7e1b50f7eba9c43106c84cb5663b014133d5dbe5bedc. Candidate count: 256.

Survivor count: 1.

37. Galaxy identity

Claim: SFT-ASTRO-GALAXY-001 **Family:** `galaxies_populations` **Statement:** A galaxy is a bound multi-component organization with environment and history retained.

Forward derivation

Carrier: `stars-gas-dust-dark-joint-carrier`. Relation: `bound-structure-and-history-relation`.

Retained record: `components-distance-frame-environment-and-time-held`. Evidence boundary:

`galaxy-observation-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,

SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-FEEDBACK-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

stars-gas-dust-dark-joint-carrier__galaxy-observation-bounded__bound-structure-and-history-relation__components-distance-frame-environment-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if morphology alone supplies identity or causal history.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.

What it means

This law makes **galaxy identity** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:acfade8fe9bf241e079008cee407de9767f5b57ad772580d98807b0432010f4b.

Independent certificate:

sha256:ed5126efdde5a95a40e1f80c63af6cbd49eb68835b69d6ac1ac89523b82974a9. Engine receipt:

sha256:39520fdbde70a6f44e6e576a76668c0770b8b6e5a092add65a5b688f55d7135c. Candidate count: 256.

Survivor count: 1.

38. Galactic structure

Claim: SFT-ASTRO-GALACTIC-STRUCTURE-001 **Family:** galaxies_populations **Statement:** Galactic structure retains components, projection, kinematics and resolution.

Forward derivation

Carrier: resolved-component-network-carrier. Relation: spatial-kinematic-component-relation. Retained record: band-resolution-orientation-distance-and-errors-held. Evidence boundary: structure-method-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-FEEDBACK-001, SFT-ASTRO-GALAXY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

resolved-component-network-carrier__structure-method-bounded__spatial-kinematic-component-relation__band-resolution-orientation-distance-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if projected light is treated as complete mass structure.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- DESI-DR1-001 - transport `failed_preserved`; snapshot `unresolved`.
- SDSS-DATA-001 - transport `captured`; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.
- NASA-IRSA-001 - transport `captured`; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.

What it means

This law makes **galactic structure** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:9605d9349b1444c9e48b28f96d49b85b9e3438654c18dbbac6c57f2b5ea21962.

Independent certificate:

sha256:dae64c6f8a5ac48a95efac6b37826303cfa8b2fad92cd619ce0a15012d104a38. Engine receipt:

sha256:8c55b5bfe11c3d144090c351c4852344af828c2ec7fffe05e207fc6d015617f5. Candidate count: 256.

Survivor count: 1.

39. Galactic dynamics

Claim: SFT-ASTRO-GALACTIC-DYNAMICS-001 **Family:** `galaxies_populations` **Statement:** Galactic dynamics retain tracers, baryons, unseen inference, geometry and alternatives.

Forward derivation

Carrier: `tracer-potential-motion-carrier`. Relation: `mass-distribution-to-motion-relation`. Retained record: `tracers-distance-inclination-baryons-dark-and-errors-held`. Evidence boundary: `dynamical-model-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-FEEDBACK-001, SFT-ASTRO-GALACTIC-STRUCTURE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

tracer-potential-motion-carrier__dynamical-model-bounded__mass-distribution-to-motion-relation__tracers-distance-inclination-baryons-dark-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a rotation curve uniquely identifies an untested mass model.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.

What it means

This law makes **galactic dynamics** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:beb7eaea295a68f5db3c1958329cf240d595329fa3e0a413b3111a71373379b6.

Independent certificate:

sha256:416dd1ee73a40efa2a237dcba4d54ca98f25e1e736357b74dc34286c90c01077. Engine receipt:

sha256:09221a0166047713578ad34a5ba48d2748ab3b857e0ac7b181766c37110d408d. Candidate count: 256.

Survivor count: 1.

40. Luminosity-rotation scaling

Claim: SFT-ASTRO-TULLY-FISHER-001 **Family:** galaxies_populations **Statement:** The pre-source rank-four relation is tested against complete source-bound galaxy rows and residuals.

Forward derivation

Carrier: galaxy-luminosity-velocity-carrier. Relation: rank-four-dilution-scaling-test. Retained record: band-distance-inclination-baryonic-choice-and-residuals-held. Evidence boundary: sample-and-regression-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
 SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
 SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
 SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-FEEDBACK-001, SFT-ASTRO-GALACTIC-DYNAMICS-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

galaxy-luminosity-velocity-carrier__sample-and-regression-bounded__rank-four-dilution-s
 caling-test__band-distance-inclination-baryonic-choice-and-residuals-held__evidence-cla
 ss-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no
 -extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if regression choice, selection or adverse rows are erased.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
 primary_archive_or_measurement_correspondence. Captured sources: 1; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- SPARC-BTFR-001 - transport captured; snapshot
 sha256 : a9b5776f6812398abc32c21c6c53e5d0bd88c22a708c9f875e49bd12a3a66f.

Quantitative record. The first preregistered all-row unweighted slope was

3.4863446052195514734204060355305148877413196602651; its deterministic delete-one range was

3.4662293923947706653191129549129334941032620165646 to

3.5477278299570478055812917404758994609387218522174, excluding four. Status:

adverse_first_protocol_preserved. It remains adverse. The separate source-authored method reports 3.85 ± 0.09 , with a stated systematic interval $3.5 - 4.0$; exact four lies at that boundary. Addendum blind status: not_blind_measurement_already_known_after_first_adverse. No first-result reclassification occurred.

What it means

This law makes **luminosity-rotation scaling** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:8fc3fa297ef292c50c016a29b21345e5598dfe499a04b6bc7c40ab19d9520cf1.

Independent certificate:

sha256:33860819d5a160017a0897edadb4dbcd725cde39e624fd82a798ba49eb5065a9. Engine receipt:

sha256:c3828b9cd507fa3d9b9abf663f4aefd6b4348b414d424262b843d05a45aa295e. Candidate count: 256.

Survivor count: 1.

41. Galaxy evolution

Claim: SFT-ASTRO-GALAXY-EVOLUTION-001 **Family:** galaxies_populations **Statement:** Galaxy evolution distinguishes population sequence from individual causal history.

Forward derivation

Carrier: time-ordered-galaxy-population-carrier. Relation: assembly-transformation-quenching-relation. Retained record: selection-redshift-mass-environment-and-progenitors-held. Evidence boundary: population-history-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-FEEDBACK-001, SFT-ASTRO-TULLY-FISHER-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

time-ordered-galaxy-population-carrier__population-history-bounded__assembly-transformation-quenching-relation__selection-redshift-mass-environment-and-progenitors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject progenitor bias or redshift selection erasure.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- DESI-DR1-001 - transport `failed_preserved`; snapshot unresolved.
- SDSS-DATA-001 - transport `captured`; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.
- NASA-IRSA-001 - transport `captured`; snapshot
sha256: 1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.

What it means

This law makes **galaxy evolution** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 8e5d877919a8e2a30ddf2314f51739628ff7874a5457d462761a2d895252c1d8.

Independent certificate:

sha256: f361fc5347efef4d11d54551aff54dcbad7d22433f97873589a7140db8fe6c9a. Engine receipt:

sha256: b8cdebfea3ab8986671702da489624ec8c20478bae24ce8bcfe4841aa4e2dd98. Candidate count: 256.

Survivor count: 1.

42. Groups, clusters and environment

Claim: SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001 **Family:** `galaxies_populations` **Statement:** Cluster inference retains membership, gas, potential, selection and mass method.

Forward derivation

Carrier: `member-galaxy-gas-potential-carrier`. Relation: `membership-binding-environment-relation`. Retained record: `selection-redshift-membership-mass-method-and-errors-held`. Evidence boundary: `cluster-definition-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,

SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-FEEDBACK-001, SFT-ASTRO-GALAXY-EVOLUTION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

member-galaxy-gas-potential-carrier__cluster-definition-bounded__membership-binding-env
 ironment-relation__selection-redshift-membership-mass-method-and-errors-held__evidence-
 class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows_
 _no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if projection or one mass proxy is treated as exact membership and mass.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
 primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- SDSS-DATA-001 - transport captured; snapshot
 sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.
- NASA-IRSA-001 - transport captured; snapshot
 sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.

What it means

This law makes **groups, clusters and environment** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:5f2062148b5b5b2f7348684026e17e8927ec35ae8207c2642a57b75e0796db71.

Independent certificate:

sha256:4feca8165d72ef8fe565f90d7ecb3524da369103dea25d5643458ac2ed160aaf. Engine receipt:

sha256:83121167316eb91e192b2ecaabdf73cee80c7db739a5cc1d3d0267f1c288483. Candidate count: 256.

Survivor count: 1.

43. Compact-object identity

Claim: SFT-ASTRO-COMPACT-OBJECT-001 **Family:** compact_high_energy_multimessenger **Statement:** Compact-object identity retains source, environment, signals, method and alternatives.

Forward derivation

Carrier: source-field-matter-signal-carrier. Relation: compact-state-to-observable-relation.

Retained record: position-mass-spin-environment-method-and-errors-held. Evidence boundary: compact-inference-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

source-field-matter-signal-carrier__compact-inference-bounded__compact-state-to-observable-relation__position-mass-spin-environment-method-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if one signature uniquely fixes an unobserved interior.

- false_premise - passed True: reject the generated form lacking the complete physical carrier

- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.
- ICECUBE-PUBLIC-DATA-001 - transport captured; snapshot
sha256:8f2fb1ac5bc28957f49b097d2a17bad60452b06b3302f395afdda597db29535e.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **compact-object identity** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:5c4f0f1bd4eb68485999d5fd6c8b5235fc8a6f473f7628b2b84e8855a971de58.

Independent certificate:

sha256:008d9d8ca098986bb1731c78ae7f6f6b5d10a59648f4e42c22dac58de8b65568. Engine receipt:

sha256:1d72a9707b68e99d69093885946b890ebdb3d08dfdb8e1a9c0d166335c90ec6. Candidate count: 256.

Survivor count: 1.

44. Degenerate remnant classes

Claim: SFT-ASTRO-WHITE-DWARF-NEUTRON-STAR-001 **Family:** `compact_high_energy_multimessenger`

Statement: White-dwarf and neutron-star claims retain mass, radius, composition, rotation and field evidence.

Forward derivation

Carrier: `progenitor-degenerate-remnant-carrier`. Relation: `mass-composition-pressure-endpoint-relation`. Retained record: `mass-radius-composition-field-rotation-and-errors-held`. Evidence boundary: `remnant-class-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,

SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001, SFT-ASTRO-COMPACT-OBJECT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

progenitor-degenerate-remnant-carrier__remnant-class-bounded__mass-composition-pressure
-endpoint-relation__mass-radius-composition-field-rotation-and-errors-held__evidence-cl
ass-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__n
o-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if a theoretical limit replaces measurement.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256: 66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.
- ICECUBE-PUBLIC-DATA-001 - transport captured; snapshot
sha256: 8f2fb1ac5bc28957f49b097d2a17bad60452b06b3302f395afdda597db29535e.
- NASA-HEASARC-001 - transport captured; snapshot
sha256: 030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **degenerate remnant classes** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 4595030afd53e4ebc85ce9cf2cc76c3479f83f34be28b50c5a5ad78d2c5b37d3.

Independent certificate:

sha256: 3915e446e6a6828ef287771d2ad41b813f228a4166affd68f1dd0fc3bf0b9259. Engine receipt:

sha256: 8270a0c8773e17094236ae656d4872538b39c7c3153f3c9149a7cba014896dd0. Candidate count: 256.

Survivor count: 1.

45. Black-hole candidate and horizon evidence

Claim: SFT-ASTRO-BLACK-HOLE-001 **Family:** compact_high_energy_multimessenger **Statement:** A black-hole claim retains dynamical mass, compactness, environment, messenger and alternatives.

Forward derivation

Carrier: source-orbit-accretion-wave-carrier. Relation: compactness-and-horizon-consequence-relation. Retained record: mass-spin-environment-messenger-and-alternatives-held. Evidence boundary: horizon-evidence-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001, SFT-ASTRO-WHITE-DWARF-NEUTRON-STAR-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

source-orbit-accretion-wave-carrier__horizon-evidence-bounded__compactness-and-horizon-consequence-relation__mass-spin-environment-messenger-and-alternatives-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if darkness alone proves a horizon or interior ontology.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.
- ICECUBE-PUBLIC-DATA-001 - transport captured; snapshot
sha256:8f2fb1ac5bc28957f49b097d2a17bad60452b06b3302f395afdda597db29535e.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **black-hole candidate and horizon evidence** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:75f47ad8561ae08e46cf6a708882b91ae28b417818c52281154e58c1551a2872.

Independent certificate:

sha256:e56f7991938117720cfe79f8951b5fe70ef3729115c2d3bb0f7b3e8ae44c9e44. Engine receipt:

sha256:7764fe13a9fb7b3d43062fec060acf9379bf00321f0bbcab8439813fbf3ff9ec. Candidate count: 256.

Survivor count: 1.

46. High-energy transient

Claim: SFT-ASTRO-TRANSIENT-001 **Family:** compact_high_energy_multimessenger **Statement:** A transient links source, event, localization, time, bands and afterglow with alternatives.

Forward derivation

Carrier: `source-event-multiband-carrier`. Relation: `progenitor-to-transient-relation`. Retained record: `localization-time-bands-selection-afterglow-and-errors-held`. Evidence boundary: `transient-class-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001, SFT-ASTRO-BLACK-HOLE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

source-event-multiband-carrier__transient-class-bounded__progenitor-to-transient-relation__localization-time-bands-selection-afterglow-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if one light-curve feature fixes a unique progenitor.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256: 66771560db1200bc22064d0b7cbe2515a98292d28f6e664f79beaf43d8ba0382.
- ICECUBE-PUBLIC-DATA-001 - transport captured; snapshot
sha256: 8f2fb1ac5bc28957f49b097d2a17bad60452b06b3302f395afdda597db29535e.
- NASA-HEASARC-001 - transport captured; snapshot
sha256: 030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **high-energy transient** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 0694ae0d2d9438317d4d7e282e233201992c84d8ad33d70209536cb435e9c342.

Independent certificate:

sha256: ea4bc7298122fa7a03f8968a813e71bf93e5577edcef4e3d1d1da5c1b50. Engine receipt:

sha256: c5eac33d788970c7c587923786815268fcb4c635f2f2469e16d08dc378136291. Candidate count: 256.

Survivor count: 1.

47. Multi-messenger association

Claim: SFT-ASTRO-MULTIMESSENGER-001 **Family:** `compact_high_energy_multimessenger` **Statement:**

Multi-messenger association retains localization, clocks, selection and background coincidence.

Forward derivation

Carrier: photon-wave-neutrino-particle-event-carrier. Relation: space-time-source-association-relation. Retained record: localizations-clocks-selection-background-and-errors-held. Evidence boundary: association-window-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001, SFT-ASTRO-TRANSIENT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

photon-wave-neutrino-particle-event-carrier__association-window-bounded__space-time-source-association-relation__localizations-clocks-selection-background-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if temporal proximity alone proves common origin.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.

- ICECUBE-PUBLIC-001 - transport captured; snapshot
sha256:8f2fb1ac5bc28957f49b097d2a17bad60452b06b3302f395afdda597db29535e.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **multi-messenger association** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:f473826f9c3736a49058496d045ec507c3621eb84b9d0b7da72b18c1b155ccc7.

Independent certificate:

sha256:b974c2ce6fd98dd5f52076b68657d41e325d9059474d16d38f24fa3554dc7b88. Engine receipt:

sha256:98f412670daf803b95eadc7246a7a5749e56caaf065a2266f008cd4be971e921. Candidate count: 256.

Survivor count: 1.

48. Compact-object population

Claim: SFT-ASTRO-COMPACT-POPULATION-001 **Family:** compact_high_energy_multimessenger **Statement:** Compact populations retain detection selection, censoring and source history.

Forward derivation

Carrier: selection-bounded-remnant-set-carrier. Relation: formation-merger-population-relation. Retained record: survey-volume-selection-masses-spins-and-censoring-held. Evidence boundary: population-model-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001, SFT-ASTRO-MULTIMESSENGER-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

selection-bounded-remnant-set-carrier__population-model-bounded__formation-merger-population-relation__survey-volume-selection-masses-spins-and-censoring-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-e

extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if detected events are treated as an unbiased population.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256: 66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.
- ICECUBE-PUBLIC-DATA-001 - transport captured; snapshot
sha256: 8f2fb1ac5bc28957f49b097d2a17bad60452b06b3302f395afdda597db29535e.
- NASA-HEASARC-001 - transport captured; snapshot
sha256: 030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **compact-object population** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 3d09c6675f037b5b71b07caf52e448b9ba1617c4e33939c735641641b171de8a.

Independent certificate:

sha256: 36b94990d08eedf60dbeff556d0b2b5b41651291473f75f020847a9b52ca4aa6. Engine receipt:

sha256: 6b6de3d5331b1c9e197bada2a99f2fb8c4685f13c83714b7d0d178a7ecf6c17d. Candidate count: 256.

Survivor count: 1.

49. Large-scale statistical organization

Claim: SFT-ASTRO-COSMIC-HOMOGENEITY-001 **Family:** `large_scale_expansion` **Statement:** Large-scale organization is a survey- and scale-bound tracer distribution.

Forward derivation

Carrier: `survey-volume-tracer-field-carrier`. Relation: `scale-dependent-distribution-relation`.

Retained record: `selection-mask-redshift-distance-and-errors-held`. Evidence boundary:

`survey-scale-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,

SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-COMPACT-POPULATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

survey-volume-tracer-field-carrier__survey-scale-bounded__scale-dependent-distribution-relation__selection-mask-redshift-distance-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if finite-volume uniformity is asserted without scale and selection.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256: 7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **large-scale statistical organization** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and

passing the same common engine.

Machine certificate

Derivation seal: sha256 : 4046956aca5d95c6e5eea080d9c99b39412c6affdca6d0d01014d15e2ca74a2e.

Independent certificate:

sha256 : 6c7283928481632b0830011a78b1721743780d4a882ee175fc8ccb24d8f266ce. Engine receipt:

sha256 : 7e928a24126ea2671b1fddb0fd3d38e85e0fece770f5d579573517f9205b18ea. Candidate count: 256.

Survivor count: 1.

50. Cosmic-web structure

Claim: SFT-ASTRO-COSMIC-WEB-001 **Family:** large_scale_expansion **Statement:** Web structure retains tracer, scale, method, selection and redshift.

Forward derivation

Carrier: tracer-node-filament-void-carrier. Relation: adjacency-density-history-relation. Retained record: tracer-selection-scale-redshift-method-and-errors-held. Evidence boundary: web-extraction-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-COMPACT-POPULATION-001, SFT-ASTRO-COSMIC-HOMOGENEITY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

tracer-node-filament-void-carrier__web-extraction-bounded__adjacency-density-history-relation__tracer-selection-scale-redshift-method-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a visualization substitutes for a generated structure criterion.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: `false`.

- DESI-DR1-001 - transport `failed_preserved`; snapshot `unresolved`.
- NASA-LAMBDA-001 - transport `captured`; snapshot `sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17`.
- SDSS-DATA-001 - transport `captured`; snapshot `sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e`.

What it means

This law makes **cosmic-web structure** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: `sha256:b450b3498e6501f5683585ebb05f460e0cd345416fd8b93b2769e7a8df62bfdc`.

Independent certificate:

`sha256:551f3db8e5a3aa0bcfcf1b8c46c4a038029308de462545f8c2b3b9144f0c994f`. Engine receipt:

`sha256:473d2d8bf8837e40ef8530f0aeddff56aff5cfe10a3f0f59b146be1a0ea85711b`. Candidate count: 256.

Survivor count: 1.

51. Cosmic expansion record

Claim: SFT-ASTRO-EXPANSION-RECORD-001 **Family:** `large_scale_expansion` **Statement:** Expansion is a multi-probe distance-redshift record with calibration and covariance retained.

Forward derivation

Carrier: `source-redshift-distance-time-carrier`. Relation: `distance-redshift-comparison-relation`. Retained record: `standardization-calibration-selection-covariance-and-era-held`. Evidence boundary: `expansion-probe-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,

SFT-ASTRO-COMPACT-POPULATION-001, SFT-ASTRO-COSMIC-WEB-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

source-redshift-distance-time-carrier__expansion-probe-bounded__distance-redshift-comparison-relation__standardization-calibration-selection-covariance-and-era-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if one rung or probe silently sets the law.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256: 7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **cosmic expansion record** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 2726011cfa7cd209bcb43d2179aada0cf289cc6a6834d7c45dec50e9fa2a0eba.

Independent certificate:

sha256: 535c0cf9449089358a0c5656397adc601c886bb69986fca502396fb49bc2a94e. Engine receipt:

sha256: d08db815fc6f99661fd7e78f5f67a40e4734053c672d5cb5d053bde2f349bd75. Candidate count: 256.

Survivor count: 1.

52. Distance-ladder composition

Claim: SFT-ASTRO-DISTANCE-LADDER-001 **Family:** large_scale_expansion **Statement:** A distance ladder is an explicit composition of anchors and rungs with covariance.

Forward derivation

Carrier: anchor-rung-source-carrier. Relation: calibration-composition-relation. Retained record: anchors-rungs-populations-zero-points-and-covariance-held. Evidence boundary: ladder-version-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-COMPACT-POPULATION-001, SFT-ASTRO-EXPANSION-RECORD-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

anchor-rung-source-carrier__ladder-version-bounded__calibration-composition-relation__anchors-rungs-populations-zero-points-and-covariance-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a terminal value is detached from its ladder.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **distance-ladder composition** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:4310d44d9fdd0b3bc9308301455dab439637b5f2cd34b3c84a8d5f4c32444ae4.

Independent certificate:

sha256:86f1e2316b08c0b14890579c7d2d0c41c987f1690c05bfe9c946c27f1e944abe. Engine receipt:

sha256:09bb53ce938c971978590248d20c8241f162a071f7a820fa660cfebb82d2139c. Candidate count: 256.

Survivor count: 1.

53. Gravitational-lensing observation

Claim: SFT-ASTRO-LENSING-001 **Family:** large_scale_expansion **Statement:** Lensing retains source, lens, observer, geometry, mass model and degeneracies.

Forward derivation

Carrier: source-lens-observer-image-carrier. Relation: multiple-path-distortion-relation. Retained record: redshifts-geometry-selection-mass-model-and-errors-held. Evidence boundary: lensing-model-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-COMPACT-POPULATION-001, SFT-ASTRO-DISTANCE-LADDER-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

source-lens-observer-image-carrier__lensing-model-bounded__multiple-path-distortion-rel
ation__redshifts-geometry-selection-mass-model-and-errors-held__evidence-class-explicit

`__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule`

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if an image distortion uniquely identifies mass without alternatives.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256: 7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- SDSS-DATA-001 - transport captured; snapshot
sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **gravitational-lensing observation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 47285b2539d8c9304fa82644849177e726007aa2fda09074468313318d144b39.

Independent certificate:

sha256: dfecb487549ac37f21ea490b03bea477467ebff9d97a51b0b33d564fffb5a192d. Engine receipt:

sha256: aa79bb09787cd670d87fd4a7ddfadf08bba448490936a246b6e7600d46bac534. Candidate count: 256.

Survivor count: 1.

54. Expansion-history inference

Claim: SFT-ASTRO-EXPANSION-HISTORY-001 **Family:** `large_scale_expansion` **Statement:** Expansion history is an ordered reconstruction from independently retained probes.

Forward derivation

Carrier: `multi-probe-epoch-carrier`. Relation: `ordered-expansion-state-reconstruction`. Retained record: `probes-calibration-covariance-model-boundary-and-time-held`. Evidence boundary: `historical-inference-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,

SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-COMPACT-POPULATION-001, SFT-ASTRO-LENSING-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

multi-probe-epoch-carrier__historical-inference-bounded__ordered-expansion-state-reconstruction__probes-calibration-covariance-model-boundary-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a fitted curve is relabelled direct cosmic history.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
 sha256: 7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- SDSS-DATA-001 - transport captured; snapshot
 sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **expansion-history inference** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same

common engine.

Machine certificate

Derivation seal: sha256:809205f910e2bd1deacda890b50684ba5728975c3e681d4c2791fe14a621c225.

Independent certificate:

sha256:16f1fc4b89e3c87996e766c7769129fb10431fe7c567224f3a1eac31e43c07dd. Engine receipt:

sha256:f49381be08c9ea93843dd3b65e22c9cd26a9fbaa329cd6c3ab085d55503b5d36. Candidate count: 256.

Survivor count: 1.

55. Cosmic thermal history

Claim: SFT-ASTRO-THERMAL-HISTORY-001 **Family:** early_background_abundance **Statement:** Thermal history retains species, interaction, scale, temperature and evidence class by epoch.

Forward derivation

Carrier: ordered-radiation-matter-state-carrier. Relation: dilution-interaction-transition-relation. Retained record: temperature-scale-species-epoch-and-evidence-held. Evidence boundary: thermal-reconstruction-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-EXPANSION-HISTORY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

ordered-radiation-matter-state-carrier__thermal-reconstruction-bounded__dilution-interaction-transition-relation__temperature-scale-species-epoch-and-evidence-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a present background uniquely supplies all earlier states.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **cosmic thermal history** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:a62a3d56c17d8401ae5711c63d3b4765fc7549bf9686c81aebf1889bd3502738.

Independent certificate:

sha256:a14dcfd5e9ac37c8e334522382bbcb1e83cd6791740cfa12cbbce5a357f011db. Engine receipt:

sha256:c2f4e22ae5b613c94534375ee96900a2751ff1cfc8f8ea35c3ae74c004907018. Candidate count: 256.

Survivor count: 1.

56. Cosmic microwave background

Claim: SFT-ASTRO-CMB-001 **Family:** `early_background_abundance` **Statement:** CMB evidence retains frequency, calibration, beam, masks, foregrounds and covariance.

Forward derivation

Carrier: `all-sky-frequency-temperature-polarization-carrier`. Relation: `background-anisotropy-spectrum-relation`. Retained record: `frequency-calibration-mask-beam-foregrounds-and-covariance-held`. Evidence boundary: `cmb-release-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,

SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-EXPANSION-HISTORY-001, SFT-ASTRO-THERMAL-HISTORY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

all-sky-frequency-temperature-polarization-carrier__cmb-release-bounded__background-ani
sotropy-spectrum-relation__frequency-calibration-mask-beam-foregrounds-and-covariance-h
eld__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-ret
ains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if a component-separated map is called raw observation.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement /`
`primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **cosmic microwave background** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:806c1220714f451e2fc205792b13182a2a524e003b0026cce6d8e82ddb9d1e47.

Independent certificate:

sha256:f7b966300db8bc75a9a123f7e60c4045aa9db65bfff4777597df852a834c321af. Engine receipt:

sha256:7118549efc4e4f62f0a3f11bd35f87cc36682d2f557515aeea40d8b0b3d5a557. Candidate count: 256.
Survivor count: 1.

57. Background acoustic structure

Claim: SFT-ASTRO-ACOUSTIC-STRUCTURE-001 **Family:** early_background_abundance **Statement:** Acoustic structure retains harmonic features, covariance and extraction boundary.

Forward derivation

Carrier: angular-spectrum-feature-carrier. Relation: bounded-harmonic-recurrence-relation. Retained record: multipoles-peaks-covariance-foregrounds-and-model-boundary-held. Evidence boundary: spectrum-analysis-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-EXPANSION-HISTORY-001, SFT-ASTRO-CMB-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

angular-spectrum-feature-carrier__spectrum-analysis-bounded__bounded-harmonic-recurrence-relation__multipoles-peaks-covariance-foregrounds-and-model-boundary-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a selected peak or harmonic fit erases adverse multipoles.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 0. The

post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **background acoustic structure** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:929c153b2747f4c46f0795820af65f787e2489853adf87460a47e58a8dc13660.

Independent certificate:

sha256:f2657b826511392ac1acb8561885144d38e97057aefc23c0276a2c71e732749d. Engine receipt:

sha256:897cd20cee5b797d7833de676b37dbd87a27ad30282b26fc64357198366bbea4. Candidate count: 256.

Survivor count: 1.

58. Primordial abundance record

Claim: SFT-ASTRO-PRIMORDIAL-ABUNDANCE-001 **Family:** early_background_abundance **Statement:**

Primordial abundance inference retains species, sites, methods, depletion and stellar processing.

Forward derivation

Carrier: species-environment-abundance-carrier. Relation: reaction-history-to-abundance-relation. Retained record: species-sites-methods-depletion-and-errors-held. Evidence boundary: abundance-inference-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-EXPANSION-HISTORY-001, SFT-ASTRO-ACOUSTIC-STRUCTURE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

species-environment-abundance-carrier__abundance-inference-bounded__reaction-history-to-abundance-relation__species-sites-methods-depletion-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if current abundance is silently relabelled primordial.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256: 7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- NASA-HEASARC-001 - transport captured; snapshot
sha256: 030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **primordial abundance record** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: de4937f016e57a3b955b8eed89ed8e61f3b53a3365954fc1aaab4557e19aba5c.

Independent certificate:

sha256: 3025487a0b98dc72a526d2c12080f7db19a78f5b6429e320f50ac4997579a654. Engine receipt:

sha256: 7cf55eed144d2fad2357efe5c54fdc92510151d15e73c07321723395df4883fb. Candidate count: 256.

Survivor count: 1.

59. Recombination and photon release

Claim: SFT-ASTRO-RECOMBINATION-001 **Family:** early_background_abundance **Statement:** Recombination links matter state, opacity and photon release with evidence boundaries.

Forward derivation

Carrier: matter-radiation-interaction-state-carrier. Relation: binding-opacity-release-relation. Retained record: species-temperature-redshift-optical-depth-and-evidence-held. Evidence boundary: recombination-model-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,

SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
 SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-EXPANSION-HISTORY-001, SFT-ASTRO-PRIMORDIAL-ABUNDANCE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

matter-radiation-interaction-state-carrier__recombination-model-bounded__binding-opacity-release-relation__species-temperature-redshift-optical-depth-and-evidence-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a model epoch is relabelled direct observation.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **recombination and photon release** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and

passing the same common engine.

Machine certificate

Derivation seal: sha256 : 732c4d0c5825a9b1afbba6ebac6b66396893a90dcfef87a002ecc1e4856d74bc9.

Independent certificate:

sha256 : 0d1fdbce9bcddd4fdd39dac1d14277979597e9c685886c0ca2483b7d36b0acd0. Engine receipt:

sha256 : 55693de3cc33c46c54235ea2ed160fb034a7bf81a5af2ce2ad5a6c5f7d709843. Candidate count: 256.

Survivor count: 1.

60. Early perturbation record

Claim: SFT-ASTRO-INITIAL-PERTURBATION-001 **Family:** early_background_abundance **Statement:** An early perturbation record retains scale, spectrum, covariance and transfer to later tracers.

Forward derivation

Carrier: background-anisotropy-seed-carrier. Relation:

initial-distinction-to-later-structure-relation. Retained record:

scale-spectrum-phase-covariance-and-transfer-held. Evidence boundary: seed-inference-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-EXPANSION-HISTORY-001, SFT-ASTRO-RECOMBINATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

background-anisotropy-seed-carrier__seed-inference-bounded__initial-distinction-to-late
r-structure-relation__scale-spectrum-phase-covariance-and-transfer-held__evidence-class
-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-e
xtra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if later structure uniquely reconstructs an unrecorded initial state.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **early perturbation record** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:d8e02965bdf73aa4bc70a8ef1603beeeb4b5f6d66b98b6044c7a2e0c5f490238.

Independent certificate:

sha256:abed46a2470051f839e8ddd1bb8aa85914f6eacfdcc77250ed2a7f107bb4d7ea. Engine receipt:

sha256:2cfecdf1565cc5f5bcf85154beb1e17120cbe8188b01418c605e77934803ac2f. Candidate count: 256.

Survivor count: 1.

61. Dark-matter evidence class

Claim: SFT-ASTRO-DARK-MATTER-EVIDENCE-001 **Family:** `dark_horizon_unobserved` **Statement:** Dark-matter evidence is a cross-probe residual with baryons and alternatives retained.

Forward derivation

Carrier: `visible-tracer-gravitational-residual-carrier`. Relation:

`observed-motion-lensing-growth-comparison`. Retained record:

`baryons-geometry-selection-alternatives-and-errors-held`. Evidence boundary:

`dark-inference-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,

SFT-ASTRO-INITIAL-PERTURBATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

visible-tracer-gravitational-residual-carrier__dark-inference-bounded__observed-motion-lensing-growth-comparison__baryons-geometry-selection-alternatives-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if one anomaly or prior preference uniquely selects a carrier.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256: 7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256: 66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.

What it means

This law makes **dark-matter evidence class** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 2489c6ef560a7807256e6bc9797a24c95adb8e9a0cd2dde0e1e0d748d0cbe698.

Independent certificate:

sha256: add825a43217ee256a5214aaf2d6443b2d069d38ca9064a2d0a999a2f96abee. Engine receipt:

sha256: da36b528cee229d263a4537a6886b66fb0c5e7448bf0f8d08e32158f31945981. Candidate count: 256.

Survivor count: 1.

62. Dark-energy evidence class

Claim: SFT-ASTRO-DARK-ENERGY-EVIDENCE-001 **Family:** dark_horizon_unobserved **Statement:** Dark-energy evidence retains distance, growth, calibration, covariance and alternatives.

Forward derivation

Carrier: distance-growth-expansion-residual-carrier. Relation: multi-probe-acceleration-comparison. Retained record: calibration-covariance-model-boundary-and-alternatives-held. Evidence boundary: dark-energy-inference-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-INITIAL-PERTURBATION-001, SFT-ASTRO-DARK-MATTER-EVIDENCE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

distance-growth-expansion-residual-carrier__dark-energy-inference-bounded__multi-probe-acceleration-comparison__calibration-covariance-model-boundary-and-alternatives-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if acceleration data alone fixes ontology or an equation of state.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.

What it means

This law makes **dark-energy evidence class** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:51b09808f74a77e4c89c9e499ef64d5a495646bd37de06c75657990b74c91b92.

Independent certificate:

sha256:a951e538d6f141760d4c5050ee9281147638ab45fdaa855a3b214eb333a2f716. Engine receipt:

sha256:c65dbbb54d1c96f9c2c02534f42eac5c4523f3c8ef09256f33013b3764437f8f. Candidate count: 256.

Survivor count: 1.

63. Cosmic horizon

Claim: SFT-ASTRO-COSMIC-HORIZON-001 **Family:** `dark_horizon_unobserved` **Statement:** A horizon is an observer- and epoch-bound causal access limit under admitted propagation.

Forward derivation

Carrier: `observer-signal-reach-carrier`. Relation: `finite-history-causal-access-relation`. Retained record: `observer-epoch-propagation-law-and-boundary-held`. Evidence boundary: `horizon-definition-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-INITIAL-PERTURBATION-001, SFT-ASTRO-DARK-ENERGY-EVIDENCE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

observer-signal-reach-carrier__horizon-definition-bounded__finite-history-causal-access-relation__observer-epoch-propagation-law-and-boundary-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if unobserved beyond-horizon state is relabelled observed absence.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256: 7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256: 66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.

What it means

This law makes **cosmic horizon** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 23c9f22f9583241e69a07c3ec8f611fb473a2ddd8e30194ca2fd476a6bd0c82d.

Independent certificate:

sha256: fe1e7e05abf7e5550490f8845f4bc5893b1f25ff003effd155dfdcd248308172. Engine receipt:

sha256: 973035e63119a5e728bb88f8606b633621a1f28d5c3a63a50e925504c79e42cd. Candidate count: 256.

Survivor count: 1.

64. Cosmic topology claim

Claim: SFT-ASTRO-TOPOLOGY-001 **Family:** dark_horizon_unobserved **Statement:** Topology claims require observable repeated-pattern consequences and controls.

Forward derivation

Carrier: observable-pattern-identification-carrier. Relation:

global-identification-consequence-relation. Retained record:

survey-mask-scale-pattern-controls-and-alternatives-held. Evidence boundary:

observable-domain-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
 SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
 SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
 SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-INITIAL-PERTURBATION-001, SFT-ASTRO-COSMIC-HORIZON-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

observable-pattern-identification-carrier__observable-domain-bounded__global-identification-consequence-relation__survey-mask-scale-pattern-controls-and-alternatives-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if mathematical possibility is counted as observation.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
 sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- GWOSC-PUBLIC-001 - transport captured; snapshot
 sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.

What it means

This law makes **cosmic topology claim** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:c187945ecee49a6ec909672e915d4df55cd28dd6b7e00f33b08bfe7dc198d342.

Independent certificate:

sha256:88d880d117bd5d5fde68df9e2c105680bc617dfd56c32121da5869f1b8bb3ff1. Engine receipt:

sha256:0040739a2ddeb6c99077b83d3fe789f427c38f159274e8f9495274ee1b7fa0e7. Candidate count: 256.

Survivor count: 1.

65. Multiverse admissibility boundary

Claim: SFT-ASTRO-MULTIVERSE-001 **Family:** dark_horizon_unobserved **Statement:** A multiverse claim enters empirical science only through a distinct observable consequence sealed before data.

Forward derivation

Carrier: declared-domain-hypothesis-carrier. Relation: observable-consequence-or-nonadmission-relation. Retained record: premises-domain-predictions-controls-and-access-held. Evidence boundary: observable-consequence-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-INITIAL-PERTURBATION-001, SFT-ASTRO-TOPOLOGY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

declared-domain-hypothesis-carrier__observable-consequence-bounded__observable-consequence-or-nonadmission-relation__premises-domain-predictions-controls-and-access-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a

fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if unobservability, anthropic selection or possibility is treated as confirmation.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256: 7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256: 66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.

What it means

This law makes **multiverse admissibility boundary** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: cb6024d3df8c3f3cdd5558351a5283c8e26100fb231e81bd9e988bc9f41c0db8.

Independent certificate:

sha256: fe07573fb0b37b59a2c944112e25e5570432212c1ecb52b0827cdc69634d413f. Engine receipt:

sha256: a117824bcf01755b83dc742009d2aad4339370fc5bf9c9626a1d3b8c7e494b6c. Candidate count: 256.

Survivor count: 1.

66. Cosmic ultimate-fate inference

Claim: SFT-ASTRO-ULTIMATE-FATE-001 **Family:** `dark_horizon_unobserved` **Statement:** Ultimate fate is conditional continuation from current records and admitted laws, not an observed terminal state.

Forward derivation

Carrier: `current-state-law-history-carrier`. Relation: `conditional-future-continuation-relation`.

Retained record: `state-laws-uncertainty-alternatives-and-domain-held`. Evidence boundary: `future-inference-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,

SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-INITIAL-PERTURBATION-001, SFT-ASTRO-MULTIVERSE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

current-state-law-history-carrier__future-inference-bounded__conditional-future-continuation-relation__state-laws-uncertainty-alternatives-and-domain-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if extrapolation uncertainty or alternative lawful continuation is erased.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-LAMBDA-001 - transport captured; snapshot
`sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17`.
- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- GWOSC-PUBLIC-001 - transport captured; snapshot
`sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382`.

What it means

This law makes **cosmic ultimate-fate inference** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:74b97003c83b3c9ac71d4b1228378463906607d074c568859bd5ac52204f73eb.

Independent certificate:

sha256:7719f41c318f8ac503948dc16e47e28663521c1954a3d6e1c20f61453e704a87. Engine receipt:

sha256:49730cfa0cd10c71cf80dd286733833e51dee356deca785b1fa0ee10e82867f6. Candidate count: 256.

Survivor count: 1.

67. Universal law and cosmic history distinction

Claim: SFT-ASTRO-LAW-HISTORY-001 **Family:** inference_prediction_handoffs **Statement:** Universal law, contingent initial condition and source-dated history remain separate.

Forward derivation

Carrier: law-initial-condition-observation-carrier. Relation: lawful-history-generation-relation. Retained record: law-receipts-initial-records-and-observations-held. Evidence boundary: ownership-boundary-declared.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ULTIMATE-FATE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

law-initial-condition-observation-carrier__ownership-boundary-declared__lawful-history-generation-relation__law-receipts-initial-records-and-observations-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a catalogue regularity is promoted to premise-free law.

- false_premise - passed True: reject the generated form lacking the complete physical carrier

- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-ADS-001 - transport `failed_preserved`; snapshot `unresolved`.
- NASA-LAMBDA-001 - transport `captured`; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- STSCI-MAST-001 - transport `captured`; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.

What it means

This law makes **universal law and cosmic history distinction** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:7615b676bcb1a70d00b85d9e6451ee6b15cecfde0efdc03c848bee376d75fe78.

Independent certificate:

sha256:c7410f1f7c30b2f43c2abe6ac06c3468d2992b359072531fe29f36e5e5d00736. Engine receipt:

sha256:c3a3d30c99d99d46c7a26df839bec5d9cd41351cd48e5ab40c5ed9cb0bb58796. Candidate count: 256.

Survivor count: 1.

68. Astronomical model-observation boundary

Claim: SFT-ASTRO-MODEL-OBSERVATION-001 **Family:** `inference_prediction_handoffs` **Statement:** A model output remains a conditional result compared with, not substituted for, observation.

Forward derivation

Carrier: `law-state-instrument-model-output-carrier`. Relation: `conditional-simulation-comparison-relation`. Retained record: `inputs-parameters-observations-errors-and-output-held`. Evidence boundary: `model-purpose-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,

SFT-ASTRO-ULTIMATE-FATE-001, SFT-ASTRO-LAW-HISTORY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

law-state-instrument-model-output-carrier__model-purpose-bounded__conditional-simulation-comparison-relation__inputs-parameters-observations-errors-and-output-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject opaque prediction or parameter fitting as derivation.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-ADS-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256: 7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.

What it means

This law makes **astronomical model-observation boundary** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: d0cc6db940af599f3d199b2ec899c3d45940fabdf814ca78afbb9430cfce5a44.

Independent certificate:

sha256: f7a47e8727ea6a1d74056fbc7f7882bcc8b617d9048b2e7fa8741a703c5ac61cc. Engine receipt:

sha256: 887f1ed59a17938a9603af5790b6e13dbcb1c63cdd768242a0a89dcfb19b8991. Candidate count: 256.

Survivor count: 1.

69. Standing astronomical prediction

Claim: SFT-ASTRO-STANDING-PREDICTION-001 **Family:** inference_prediction_handoffs **Statement:** A standing prediction is sealed before outcome and retains every adverse and missing row.

Forward derivation

Carrier: sealed-premise-prediction-outcome-carrier. Relation: predeclared-comparison-relation. Retained record: targets-metrics-baselines-adverse-and-missing-rows-held. Evidence boundary: prediction-question-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ULTIMATE-FATE-001, SFT-ASTRO-MODEL-OBSERVATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

sealed-premise-prediction-outcome-carrier__prediction-question-bounded__predeclared-comparison-relation__targets-metrics-baselines-adverse-and-missing-rows-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject post-outcome target changes or favorable-row selection.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The

post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-ADS-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.

What it means

This law makes **standing astronomical prediction** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:270b6510c1fc80f7274e0323530b0e1176f983d57cfc20812445548ccc6c3f0f.

Independent certificate:

sha256:8b7b81836bef277ed91a661cd5dfab965e26cef89cda75794e5914a0c501de43. Engine receipt:

sha256:b83b752df3a1e802277ac2e5eaf8205baeefbd00d827236838e8a57ccaa1b93. Candidate count: 256.

Survivor count: 1.

70. Unexplained astronomical observation

Claim: SFT-ASTRO-UNEXPLAINED-001 **Family:** inference_prediction_handoffs **Statement:** Unexplained means a source-bound residual after declared controls, not evidence for a preferred unknown cause.

Forward derivation

Carrier: observation-controls-alternative-carrier. Relation: residual-after-tested-explanations-relation. Retained record: data-method-controls-alternatives-and-failures-held. Evidence boundary: residual-evidence-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ULTIMATE-FATE-001, SFT-ASTRO-STANDING-PREDICTION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

observation-controls-alternative-carrier__residual-evidence-bounded__residual-after-tested-explanations-relation__data-method-controls-alternatives-and-failures-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject argument from ignorance or hidden adverse controls.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-ADS-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256: 7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.

What it means

This law makes **unexplained astronomical observation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: 4c5f2f4efe22a0ef5876fe979e0725066510a3eff19d9d55371181f8dbf5a13e.

Independent certificate:

sha256: 3d409d8701b71cd241505f4ae9157d5bc38c0d8e4fa2fa8bc5d3ab007651118b. Engine receipt:

sha256: 7af6264c33e27e20954db0642d0051f8e9b14af8664c8446daf278ed238bbcf2. Candidate count: 256.

Survivor count: 1.

71. Astrobiology and technosignature boundary

Claim: SFT-ASTRO-ASTROBIOLOGY-001 **Family:** inference_prediction_handoffs **Statement:** Life or technology claims require replicated signals and complete abiotic, instrumental and contamination controls.

Forward derivation

Carrier: environment-signal-life-hypothesis-carrier. Relation:

biosignature-or-technosignature-comparison. Retained record:

abiotic-controls-selection-contamination-and-replication-held. Evidence boundary:

life-evidence-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
 SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
 SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
 SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-ULTIMATE-FATE-001, SFT-ASTRO-UNEXPLAINED-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

environment-signal-life-hypothesis-carrier__life-evidence-bounded__biosignature-or-tech
 nosignature-comparison__abiotic-controls-selection-contamination-and-replication-held__
 evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-
 all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject an ambiguous signal or habitability proxy as life detection.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
 primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-ADS-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
 sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- STSCI-MAST-001 - transport captured; snapshot
 sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.

What it means

This law makes **astrobiology and technosignature boundary** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:1f067d4fba2910dd7062cb29edaa4072315e6ee30c5f6fa2cba36620467f79b0.

Independent certificate:

sha256:ee3c67811dbdb007ae493220f8ab7bc3b1838ad05a1d24ac4168dcbbc6658451. Engine receipt:

sha256:1b06beceb6d9a84c63a86aef6d02baf535a9244f99301d68a82778fae3d6e585. Candidate count: 256.

Survivor count: 1.

72. Astronomy scientific handoffs

Claim: SFT-ASTRO-ASTRO-HANDOFF-001 **Family:** inference_prediction_handoffs **Statement:** Astronomy consumes universal laws and exports source-dated constraints through explicit Physics, Engineering, Earth, Biology and Social boundaries.

Forward derivation

Carrier: law-instrument-earth-life-social-boundary-carrier. Relation: typed-branch-handoff-relation. Retained record: source-owner-dependency-evidence-and-use-held. Evidence boundary: branch-ownership-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ULTIMATE-FATE-001, SFT-ASTRO-ASTROBIOLOGY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

law-instrument-earth-life-social-boundary-carrier__branch-ownership-bounded__typed-branch-handoff-relation__source-owner-dependency-evidence-and-use-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a

fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if an instrument selects a law or another branch's object is silently reowned.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-ADS-001 - transport `failed_preserved`; snapshot `unresolved`.
- NASA-LAMBDA-001 - transport `captured`; snapshot
`sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17`.
- STSCI-MAST-001 - transport `captured`; snapshot
`sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701`.

What it means

This law makes **astronomy scientific handoffs** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favorable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: `sha256:c6b99ac95649612e4f1407601647e18bc33a4ca9bf09bbfbeeaa525448813f69`.

Independent certificate:

`sha256:4df89f0be221973480f47e417d1561e932f93827b5a8070ceebfe15a09e3d5dc`. Engine receipt:

`sha256:1d8b4042aa2faf71d32266ffcd173e2d2462772695741ae0f290d55d0ba6a883`. Candidate count: 256.

Survivor count: 1.

Historical reconciliation

The V1/V2 census contains 763 entries. Every entry was reviewed. The 53 Astronomy-owned questions were registered as questions before the V3 derivation and mapped to admitted present claims. Prior answers were not used as premises. Universal Physics questions were preserved as typed handoffs rather than relabelled Astronomy claims.

Limits and extension-open status

This foundation does not claim that every astronomical catalogue row has been ingested, that every contingent initial condition is forced by structure, or that every instrument systematic has vanished. It claims something more precise: the foundational forms required to state, compare and falsify astronomical claims have been generated and admitted at the declared boundary, and all known V1/V2 Astronomy questions are accountable. New observations, objects, methods and lawful derivations are invited. A new result must extend the trace; it may not rewrite failed transports, adverse rows or dependency ownership.

Full-field continuation roadmap

- complete positional and time-domain catalogues.

- complete stellar and planetary populations.
- complete interstellar and chemical-evolution observations.
- complete galaxy and large-scale surveys.
- complete compact/high-energy/multimessenger catalogues.
- complete precision cosmic-history and early-universe evidence.
- complete astrobiological and observatory handoffs.

Each extension will be decomposed into explicit obligations, sealed before outcomes, admitted claim by claim and followed by a versioned paper update. Astronomy remains current-evidence closed at its foundation and open to discovery.

Reproduction

This Astronomy closure passed six focused tests, 72 sequential engine admissions, 72 independent validators and the branch integration audit. After all fifteen foundations were integrated, the untouched repository command `python3 -m sft verify-all` passed on 28 July 2026: 1,319/1,319 derivations replayed, 969/969 unit and end-to-end tests passed, 1,264/1,264 core executable lines covered, 1,011 empirical claims audited, 114/114 formal Physics claims reaching measurement and 5/5 live exact NIST/CODATA checks. The measured runtime was 4,753.740 seconds.

Rights, participation and scientific admission

Open criticism is welcome and does not require credentials. Scientific admission requires the shared protocol. Contributors may use the Ernos Labs designation only while preserving the theorem, constraints, complete enumeration, adverse evidence, immutable engine authority, retained authorship and open licenses. A fork that weakens those conditions remains free software but is not an Ernos Labs scientific admission.

Author: Maria Smith **Email:** Maria.Smith.Sftoe@gmail.com **Discord:** <https://discord.gg/ucwGryVxGr> **GitHub:** <https://github.com/MettaMazza> **Paper license:** CC BY 4.0 **Code license:** Apache-2.0

Machine identities

- Inventory: sha256:466048a43ba8e3fc91f33f231f18571bf80a3d16e934a18ec060678e814421cd
- Prior audit: sha256:4dbda9dcdf66a0d9e23605be4bfd41e2a5c28cd221989cac6d0d75192e322463
- Integration audit: sha256:769af612334b3d79346b79430acb82b731bf8174f5cce60511773f88b991e9d4
- Source registry: sha256:b5d65c48dbbb538cb3ce95a9e57f3f7ad45d0e3c821013285a48b723a51a7126
- External target census:
sha256:76a74ee67434d0e774624ca6fdac747dc729a6578c1dfd0081c1b2636364d130
- Engine: sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a
- Verification authority: sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8